25/8/69

Pre-processing before and during training

Image data, class 2 of 2. Classical operations, normalization done without leaking, medical intensity standardization, and augmentation from flips to MixUp.

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Visual Intelligence Laboratory · Khon Kaen University

A curated compilation of openly published university teaching materials. Every section names its source; original slides are shown unmodified.

Today, in three hours

0:00–0:25	**D1 · Filtering as preprocessing.** Point ops, convolution, Gaussian vs box, sharpening restraint.	CMU 16-385, Lecture 2
0:25–0:45	**D2 · Medical classical ops.** Histograms on CT, median filtering, MRI denoising, SNR and CNR.	Bagci Lab BME 495, Lectures 3–4
0:45–1:00	**D3 · Contrast, color, crop.** Histogram equalization and CLAHE, conversions, cropping and padding.	authored
1:00–1:10	*Break*	
1:10–1:30	**E1 · Preprocessing for the model.** Zero-center, normalize, per-channel statistics, and why optimizers care.	UMich EECS 498, Lecture 10
1:30–1:50	**E2 · The rule + medical standardization.** Statistics from training data only; MRI intensity standardization.	authored + Bagci L3–4
1:50–2:05	**E3 · Transfer learning.** Pretrained backbones and the statistics they expect.	Stanford CS231n, Lecture 7
2:05–2:40	**F1–F2 · Augmentation.** Flips to MixUp; medical augmentation incl. scanner artifacts.	EECS 498 L10 + Bagci L9
2:40–3:00	**F3 · Pipelines in code.** torchvision v2, Albumentations, Keras layers, MONAI, CutMix in the loader.	authored

PART D1 · 0:00–0:25

Filtering: the classical preprocessing toolbox

Point operations, convolution, and smoothing. The same kernels you tune by hand here are what a CNN's first layer learns for itself, which is why this vocabulary matters.

SOURCE DECK

Image filtering, 16-385 Computer Vision, Fall 2023, Lecture 2 · Ioannis Gkioulekas, Carnegie Mellon University.
http://16385.courses.cs.cmu.edu/fall2023/ © CMU and credited original authors · shown unmodified · selected pages

Image filtering



16-385 Computer Vision
Fall 2023, Lecture 2

http://16385.courses.cs.cmu.edu/

Linear shift-invariant image filtering

- Replace each pixel by a *linear* combination of its neighbors (and possibly itself).
- The combination is determined by the filter's *kernel*.
- The same kernel is *shifted* to all pixel locations so that all pixels use the same linear combination of their neighbors.

Example: the box filter

- also known as the 2D rect (not rekt) filter
- also known as the square mean filter

kernel $g[\cdot,\cdot] = \frac{1}{9}$

1	1	1
1	1	1
1	1	1

ง่ายสุด
ทำให้ภาพเบลอ
แต่ขนาดภาพเท่าเดิม



- replaces pixel with local average
- has smoothing (blurring) effect

… and the result is

$g[\cdot,\cdot]$

kernel

$$\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$$

$f[\cdot,\cdot]$

image

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	90	0	90	90	90	0	0
0	0	0	90	90	90	90	90	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	90	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

$h[\cdot,\cdot]$

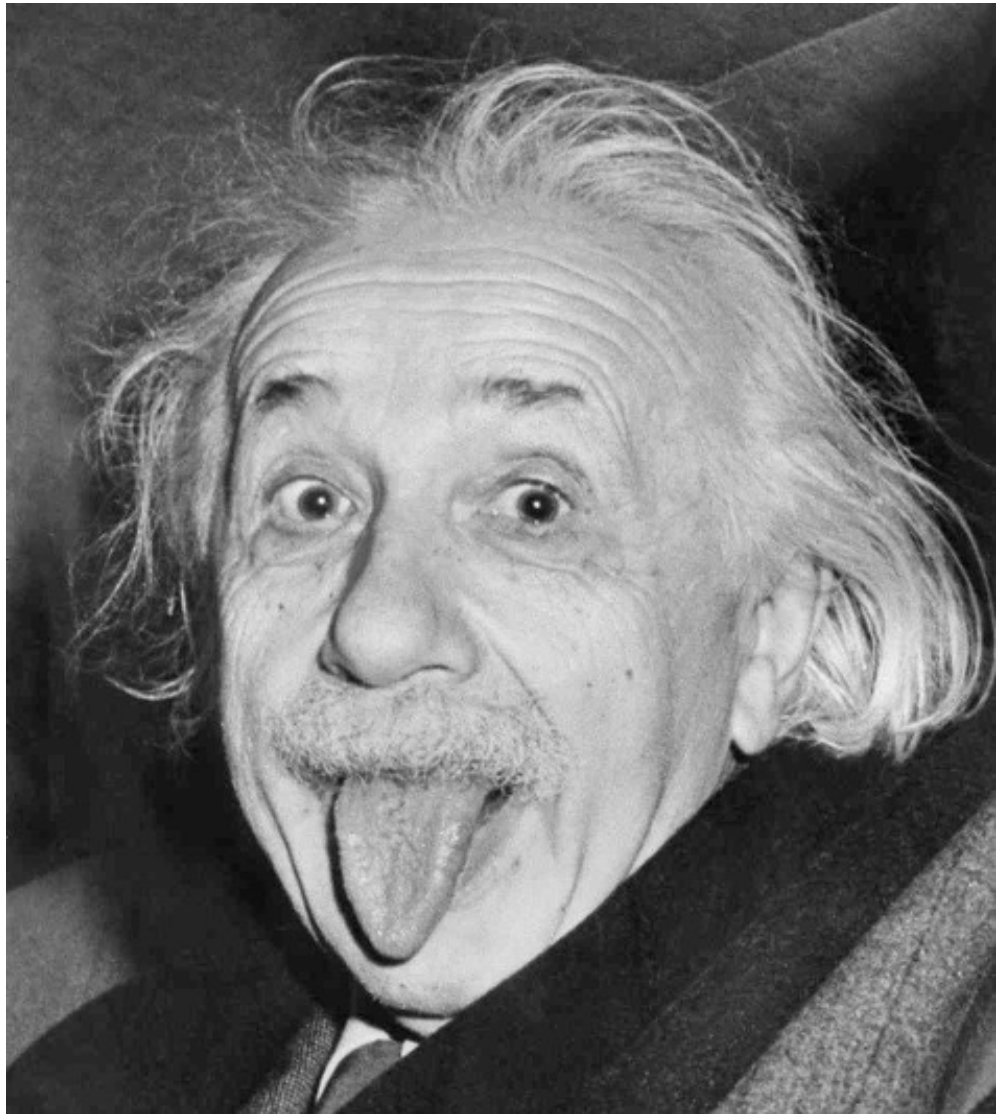
output

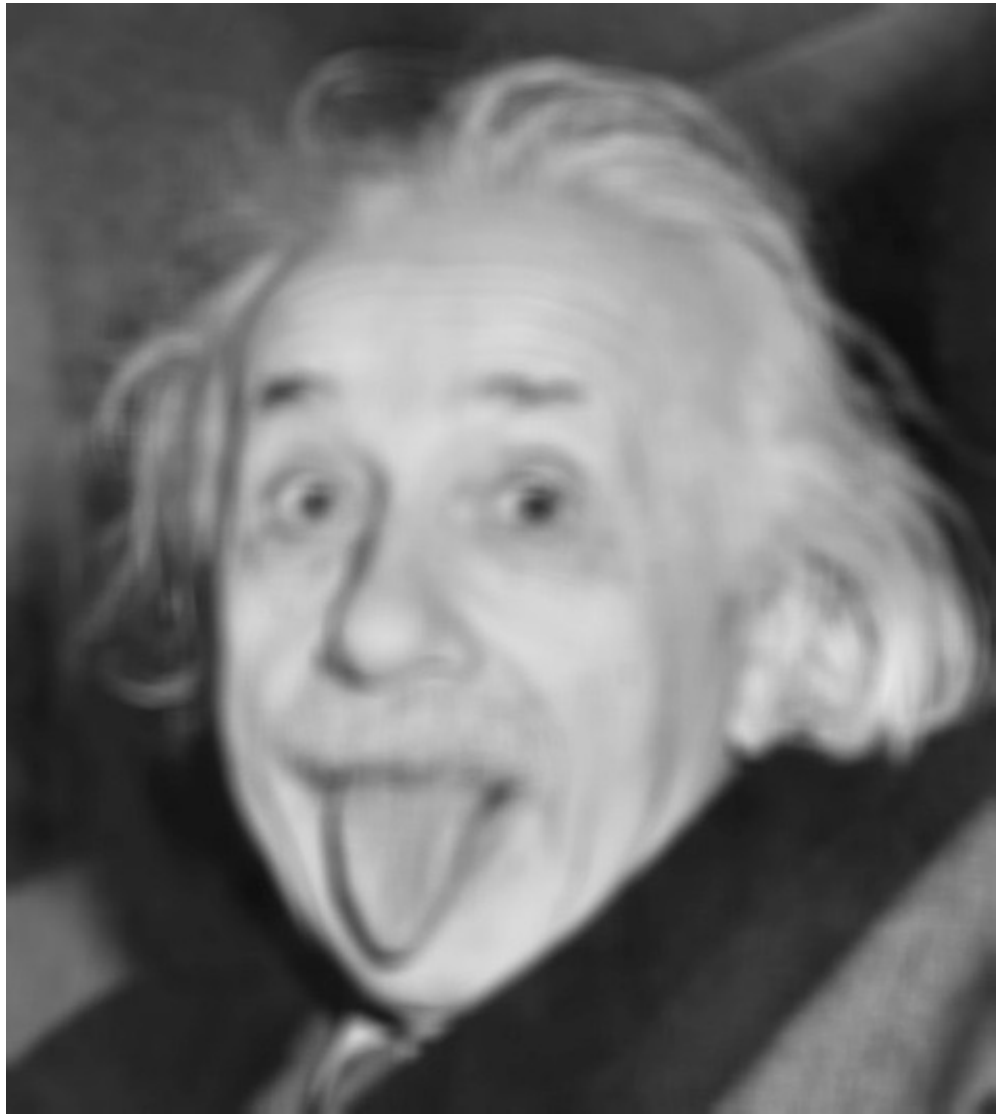
	0	10	20	30	30	30	20	10	
	0	20	40	60	60	60	40	20	
	0	30	50	80	80	90	60	30	
	0	30	50	80	80	90	60	30	
	0	20	30	50	50	60	40	20	
	0	10	20	30	30	30	20	10	
	10	10	10	10	0	0	0	0	
	10	10	10	10	0	0	0	0	

$$h[m,n] = \sum_{k,l} g[k,l]\, f[m+k, n+l]$$

output — filter — image (signal)

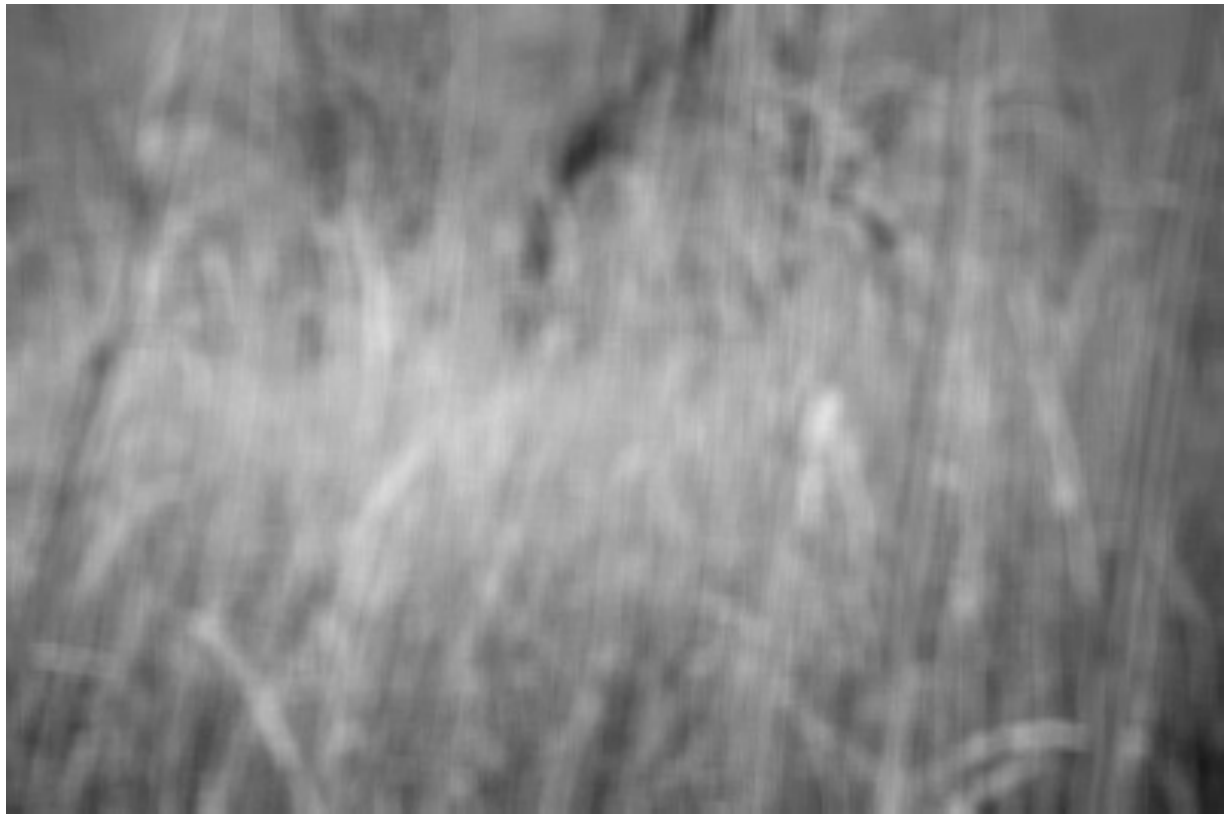
Some more realistic examples





Some more realistic examples





Some more realistic examples





The Gaussian filter

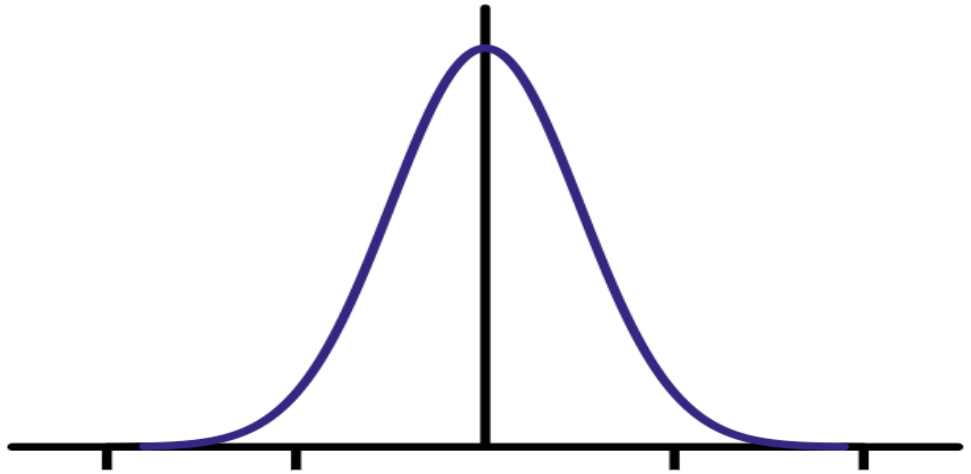
- named (like many other things) after Carl Friedrich Gauss
- kernel values sampled from the 2D Gaussian function:

$$f(i,j) = \frac{1}{2\pi\sigma^2} e^{-\frac{i^2+j^2}{2\sigma^2}}$$

- weight falls off with distance from center pixel
- theoretically infinite, in practice truncated to some maximum distance

Any heuristics for selecting where to truncate?

- usually at 2-3σ



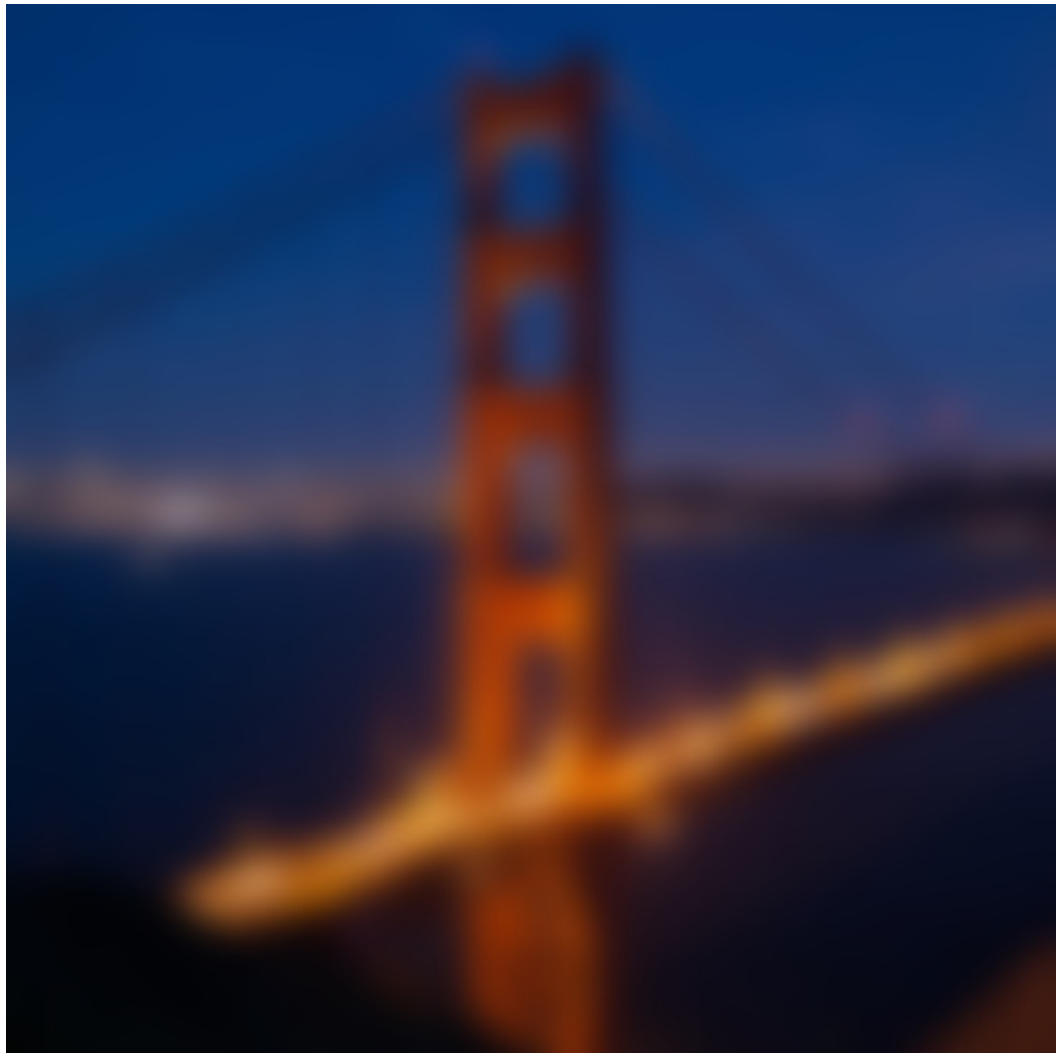
Is this a separable filter? Yes!

kernel $\frac{1}{16}$

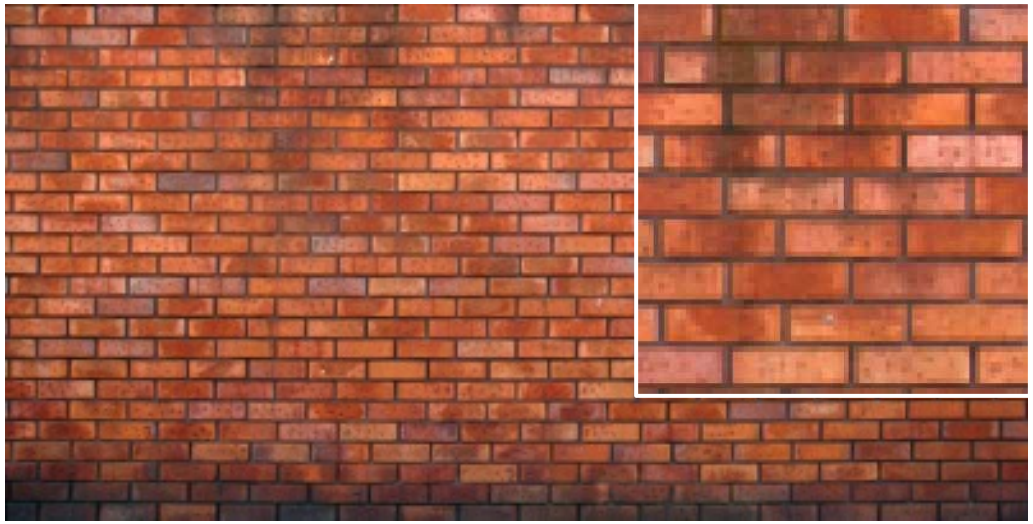
1	2	1
2	4	2
1	2	1

Gaussian filtering example

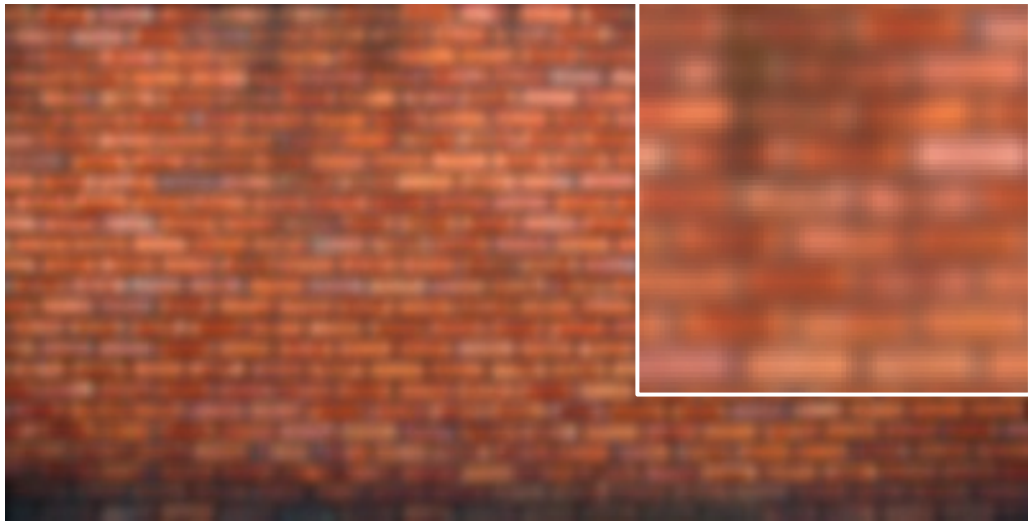




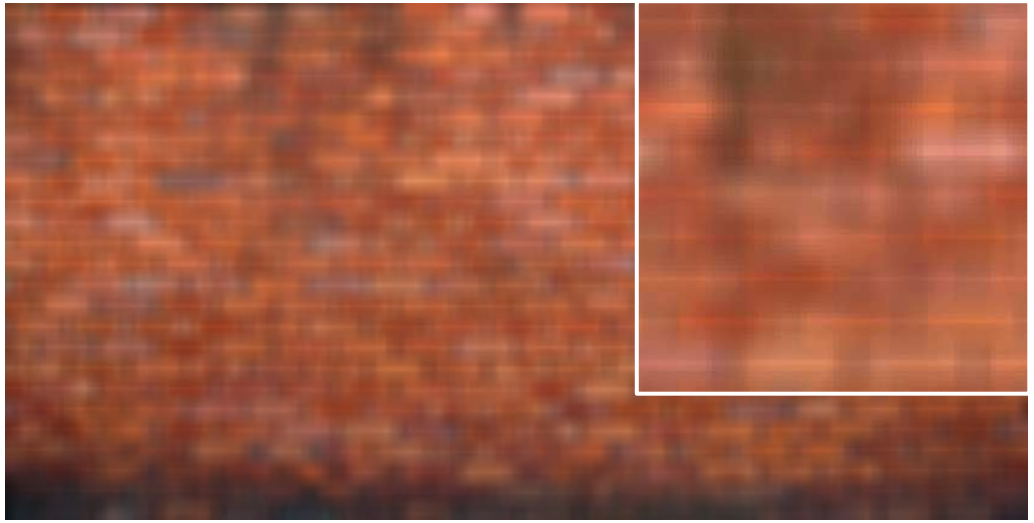
Gaussian vs box filtering



original



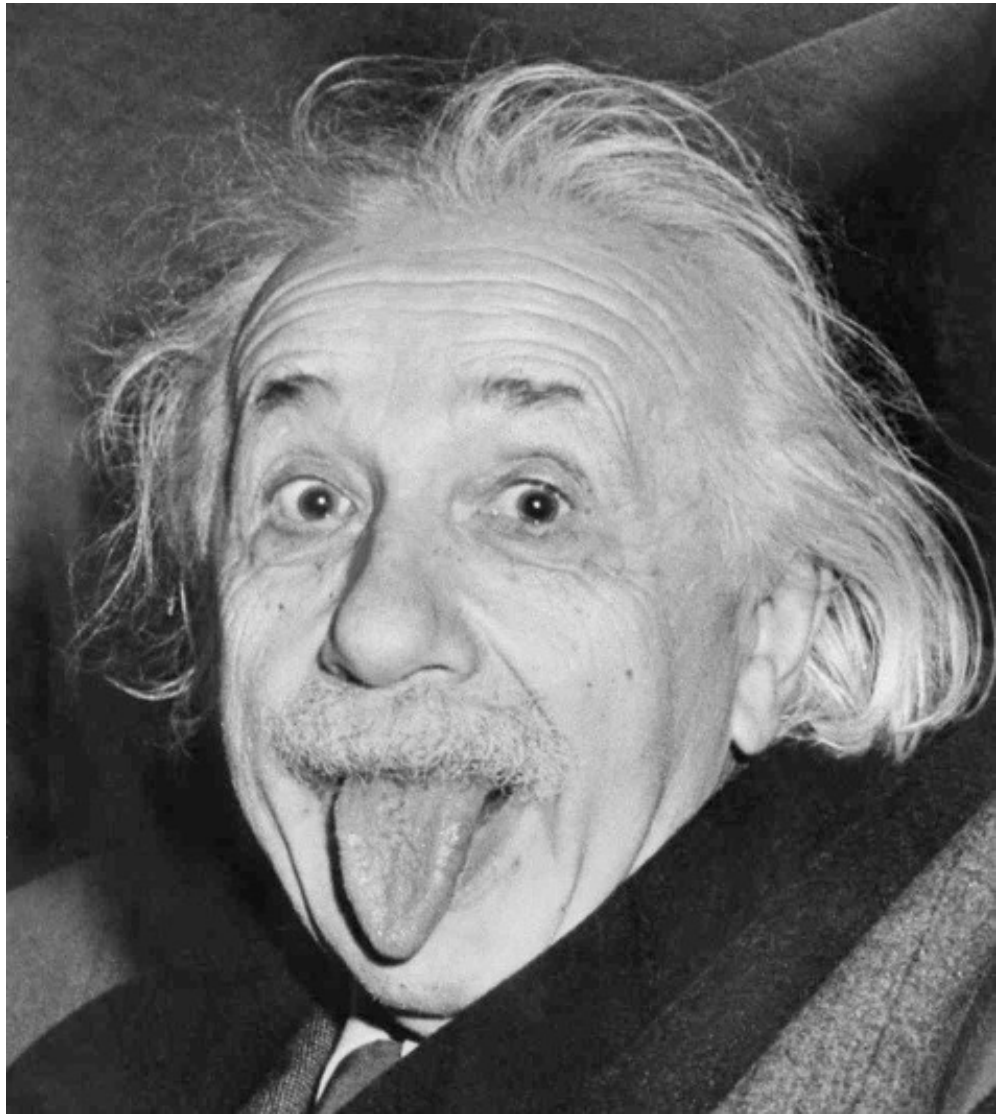
7x7 Gaussian

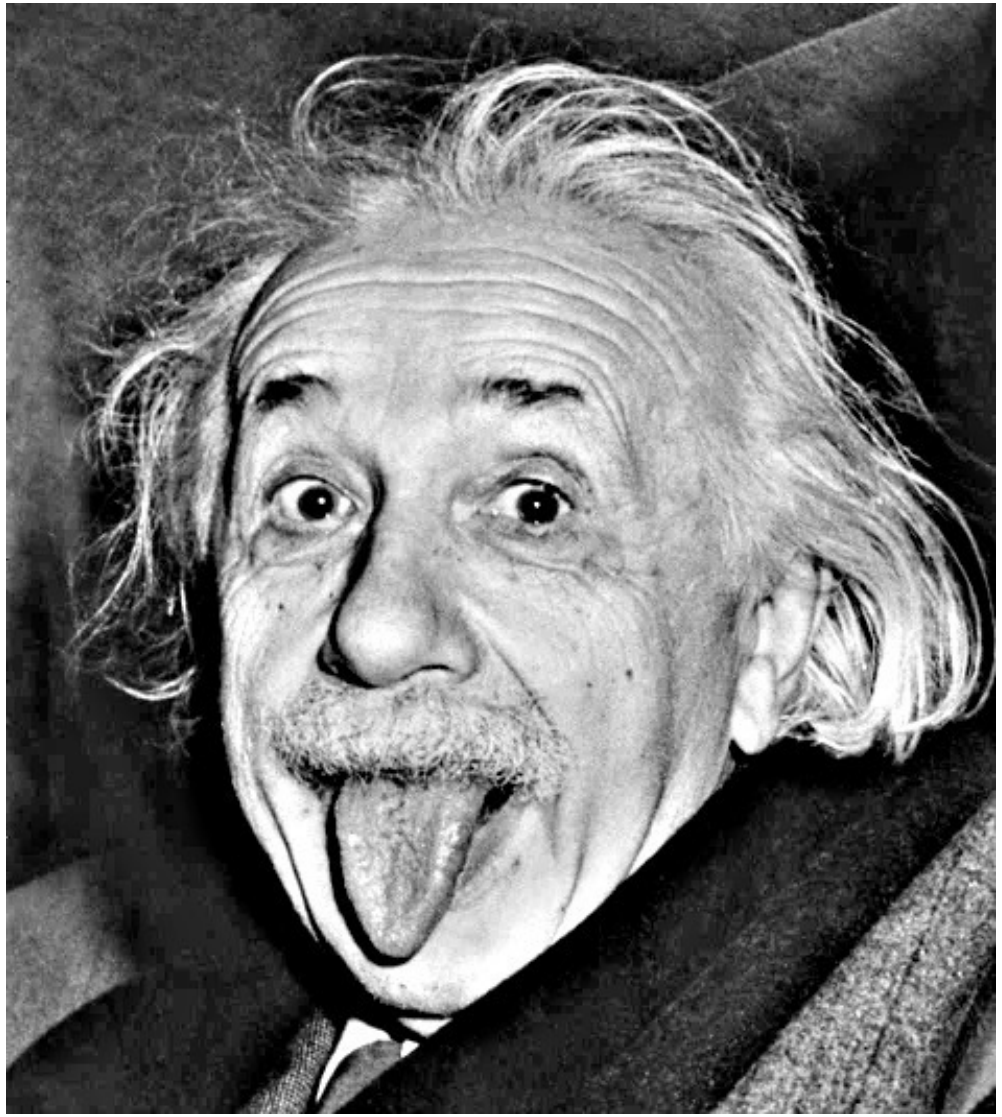


7x7 box

Which blur do you like better?

Sharpening examples





Do not overdo it with sharpening



original



sharpened



oversharpened

What is wrong in this image?

PART D2 · 0:25–0:45

The same toolbox, on medical images

Histograms that read lung disease, median filters on X-rays, MRI denoising judged by SNR and CNR: preprocessing with a clinical meaning attached to every intensity.

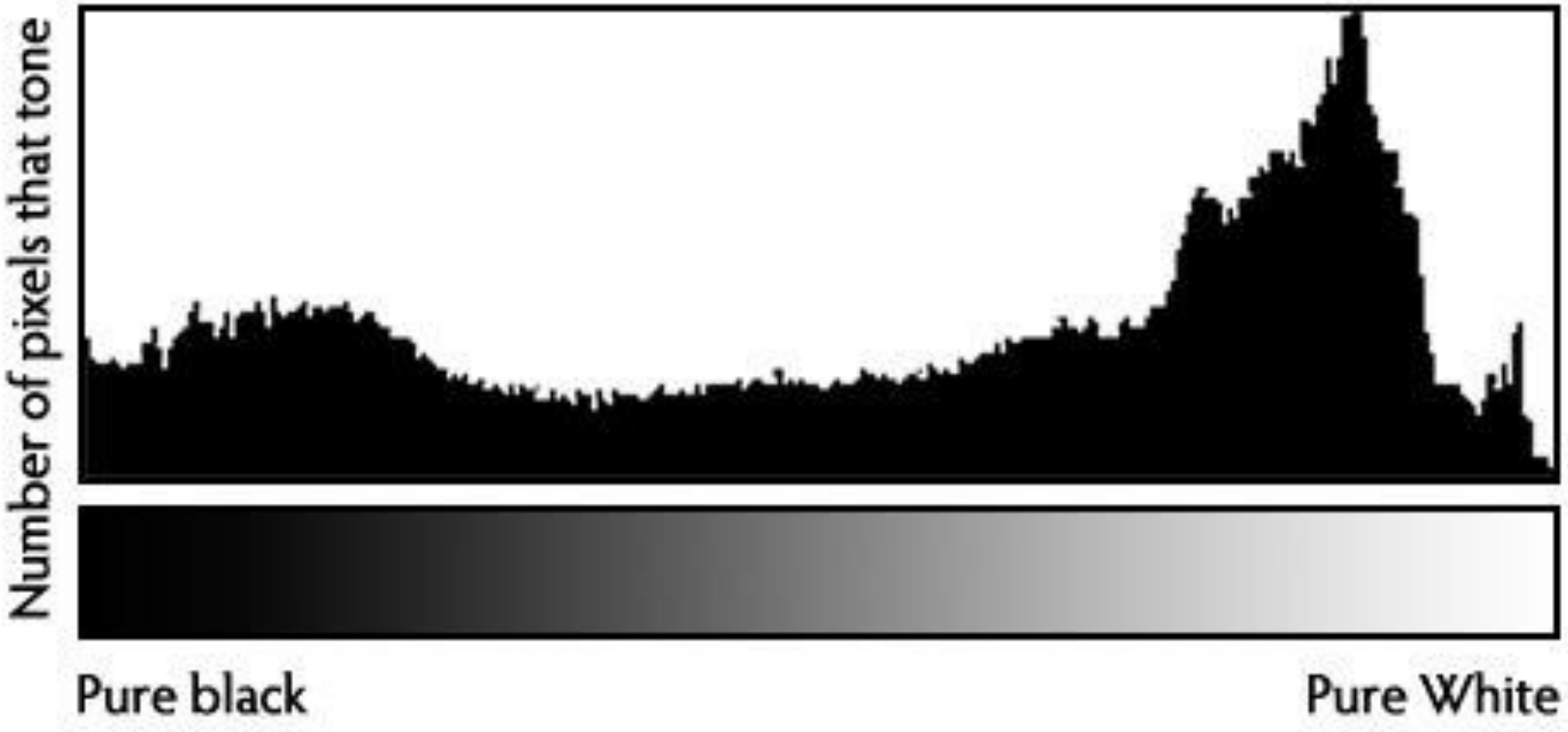
SOURCE DECK

Image preprocessing, BME 495 Deep Learning for Biomedical Imaging, Winter 2025, Lectures 3–4 · Ulas Bagci, Northwestern University.

https://bagcilab.com/courses/ publicly shared by the author · shown unmodified · selected pages

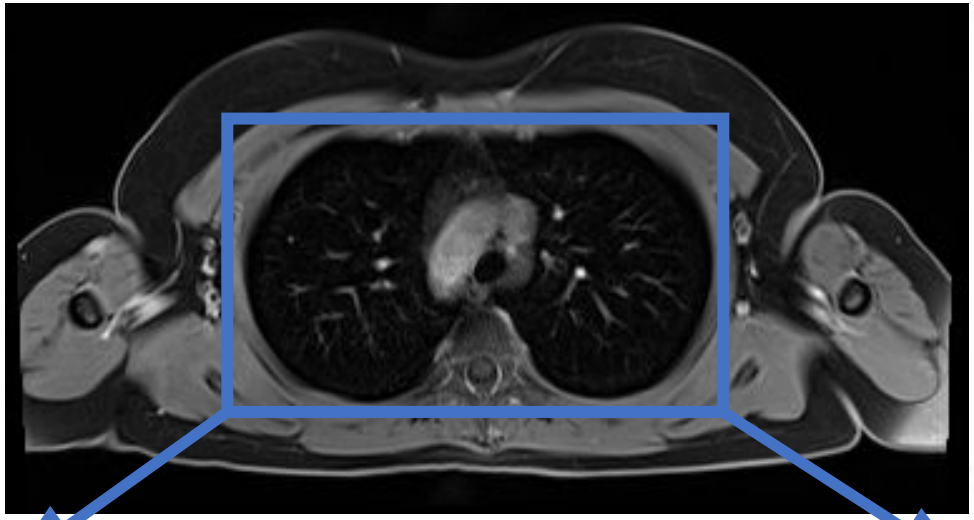

Image Histogram ?

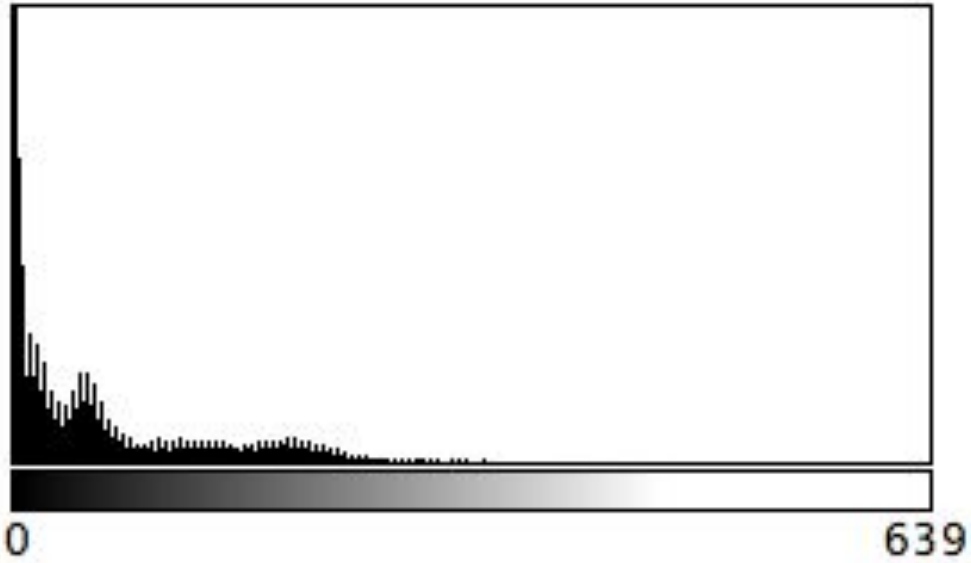
Image Histogram

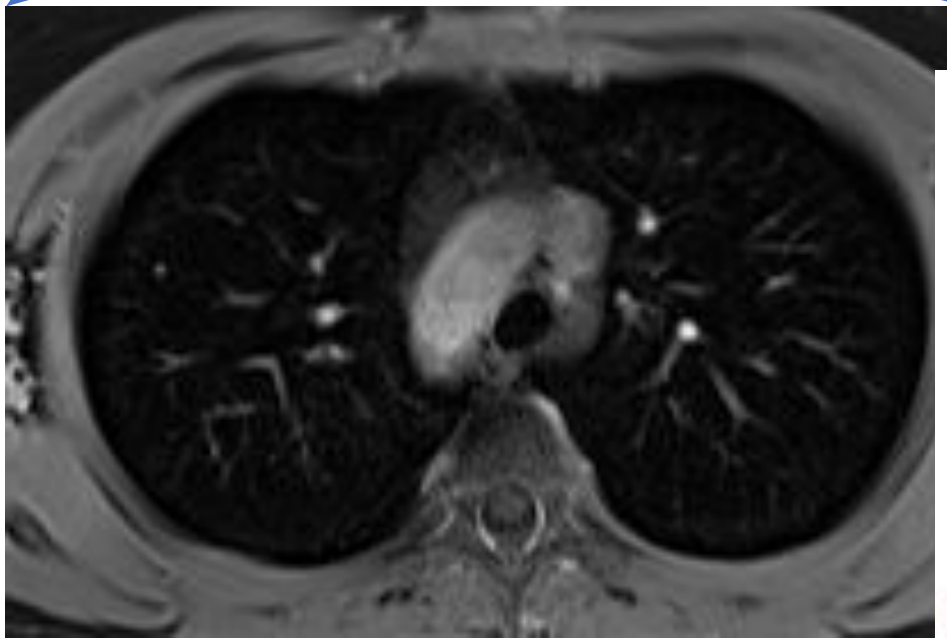


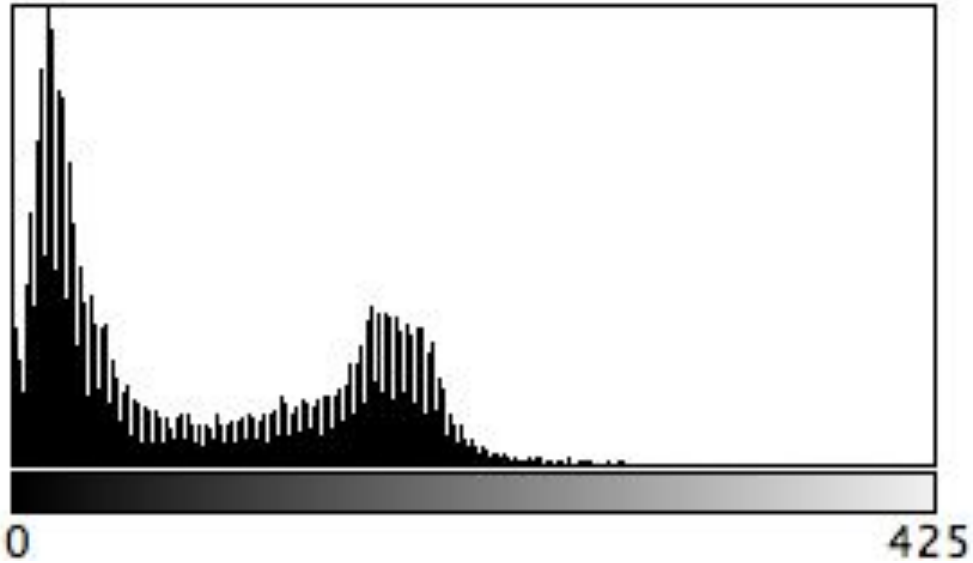
It plots the number of pixels for each intensity value.

Image Histogram – Example

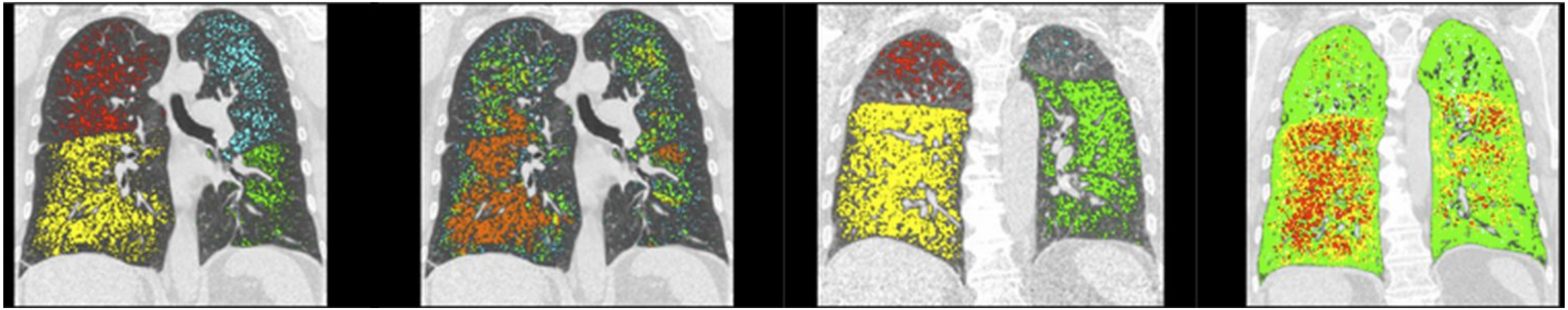


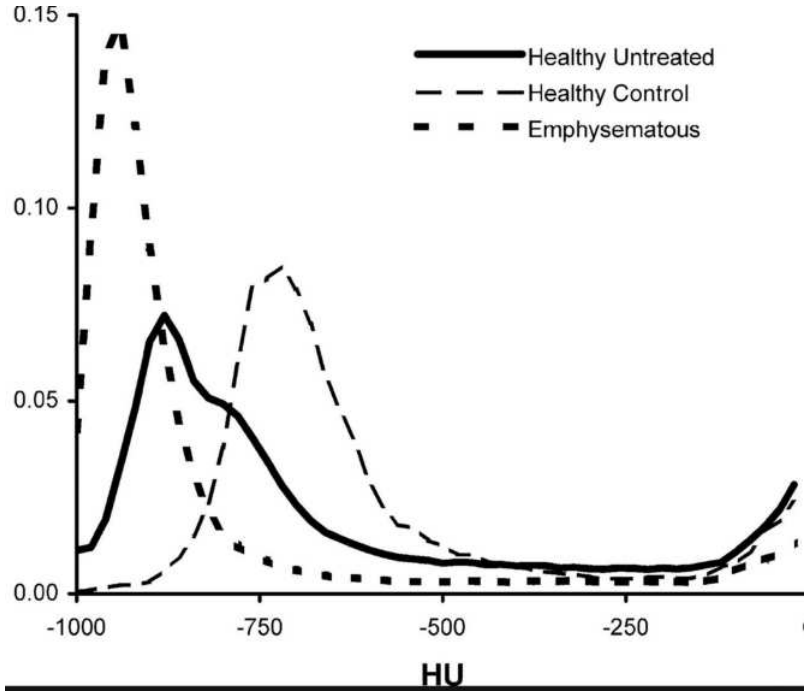






Histogram based analysis of Lung CT (credit: *Imbio*)



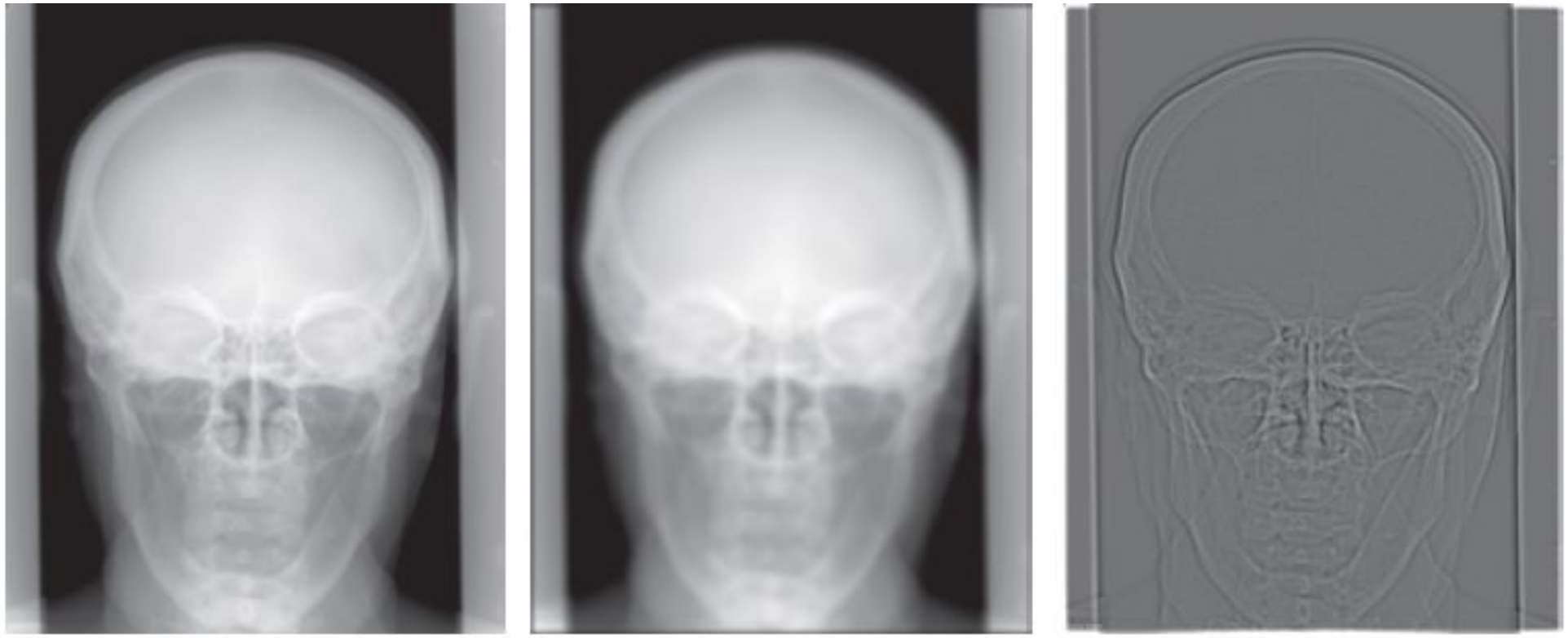


Median Filters (non-linear)

- A **Median Filter** operates over a window by selecting the median intensity in the window.

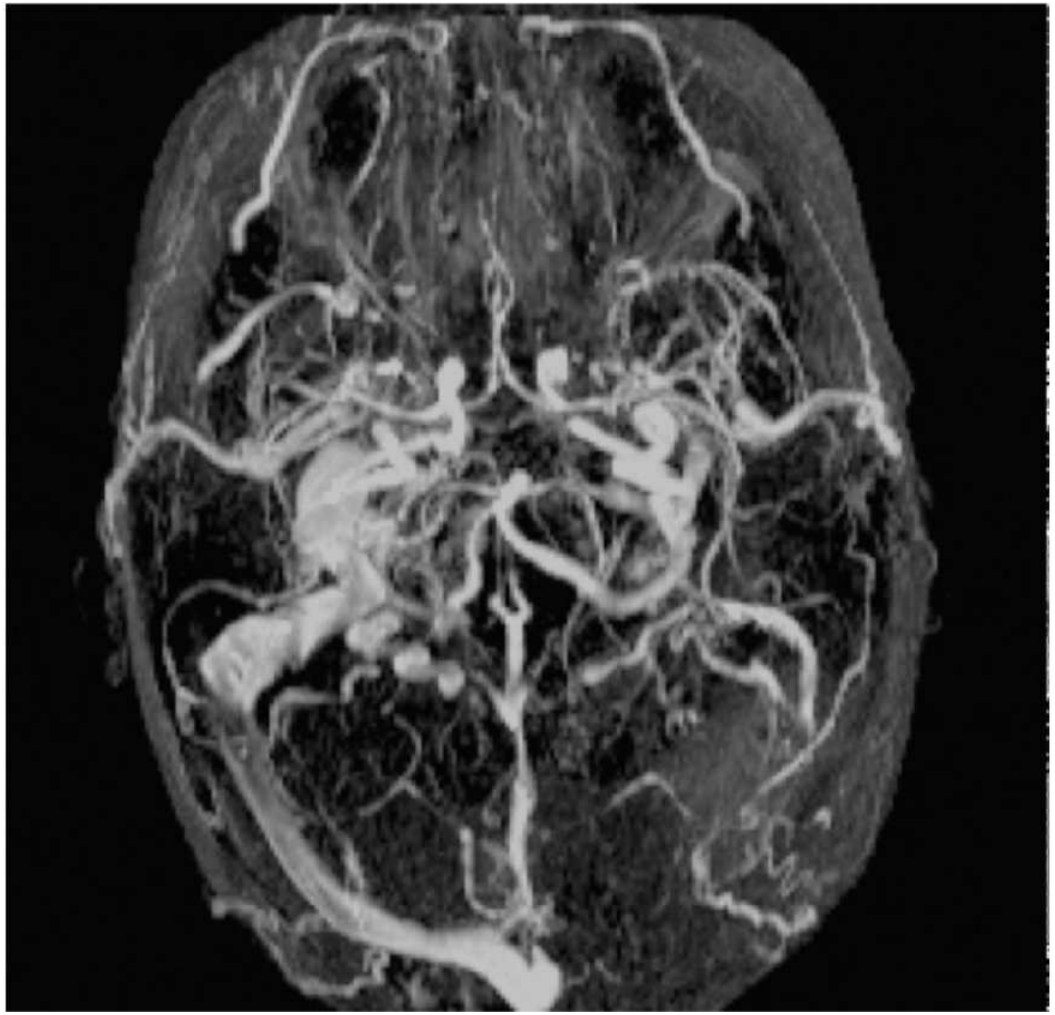


Filtering X-ray

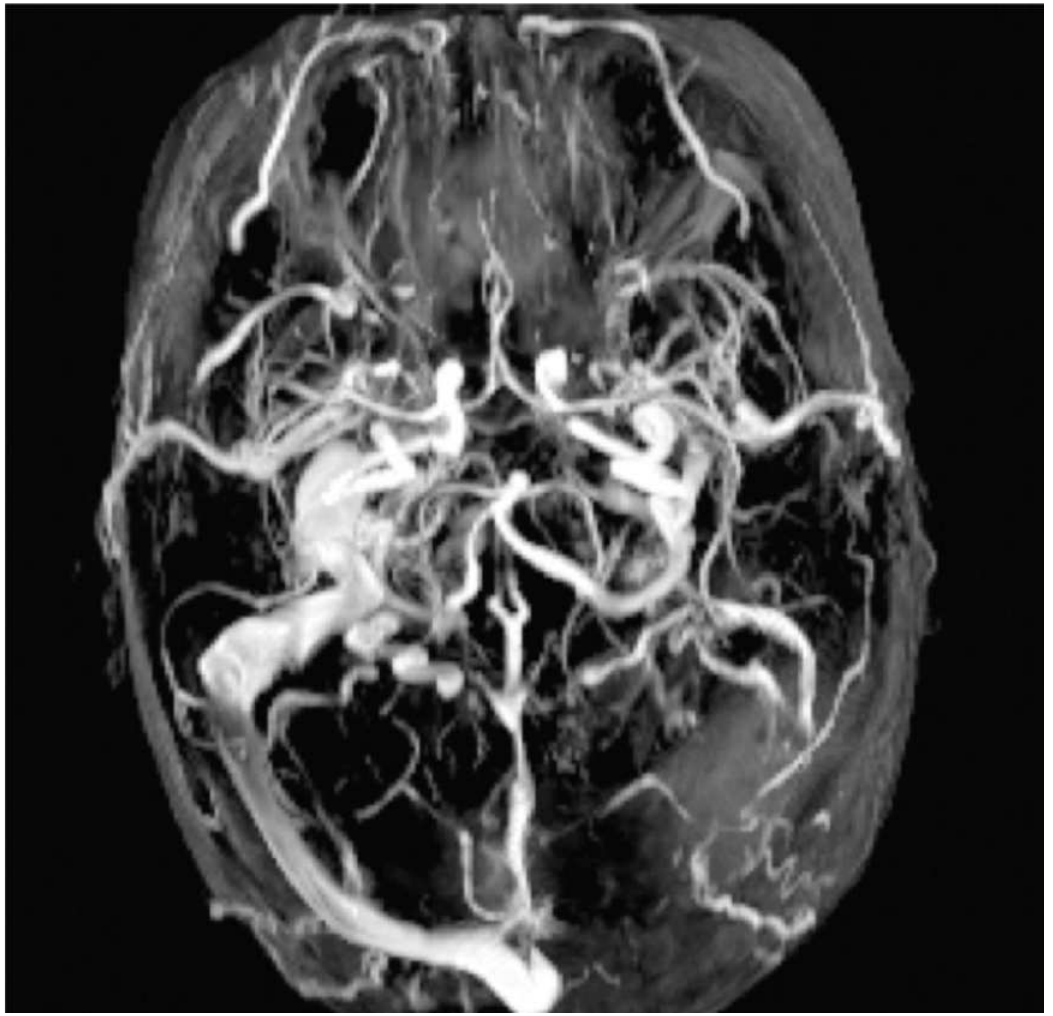


a. Radiography of the skull, b. low-pass filter with a Gaussian filter (std=15, 20 x 20), c. high-pass Filter obtained from subtracting b from a.

Adaptive Filtering: Example head MRA

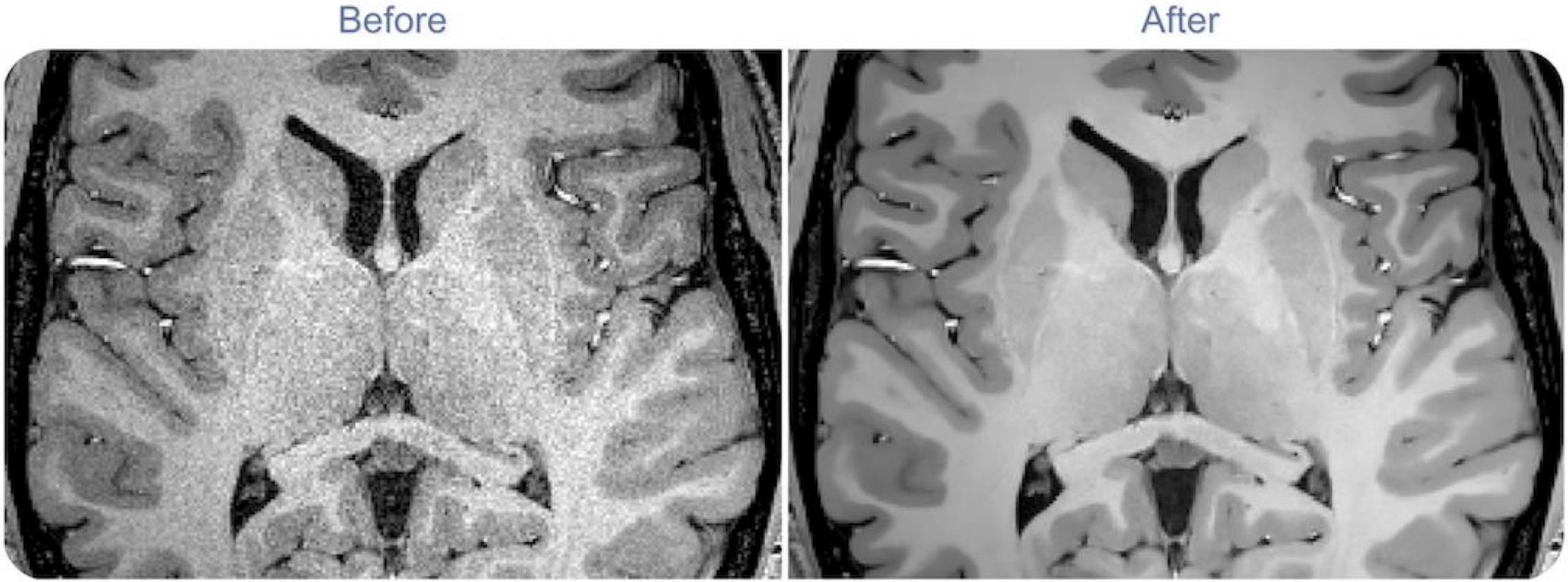


MIP of MRA data before filtering



MIP of MRA data after filtering

Smoothing MRI (Denoising)



P. Coupé et al. An Optimized Blockwise NonLocal Means Denoising Filter for 3-D Magnetic Resonance Images. IEEE TMI 2008

How to measure for evaluating noise removal algorithms?

SNR: Signal to Noise Ratio

CNR: Contrast to Noise Ratio

MRI SNR – Example 1

$$SNR = \frac{Signal}{Noise}$$

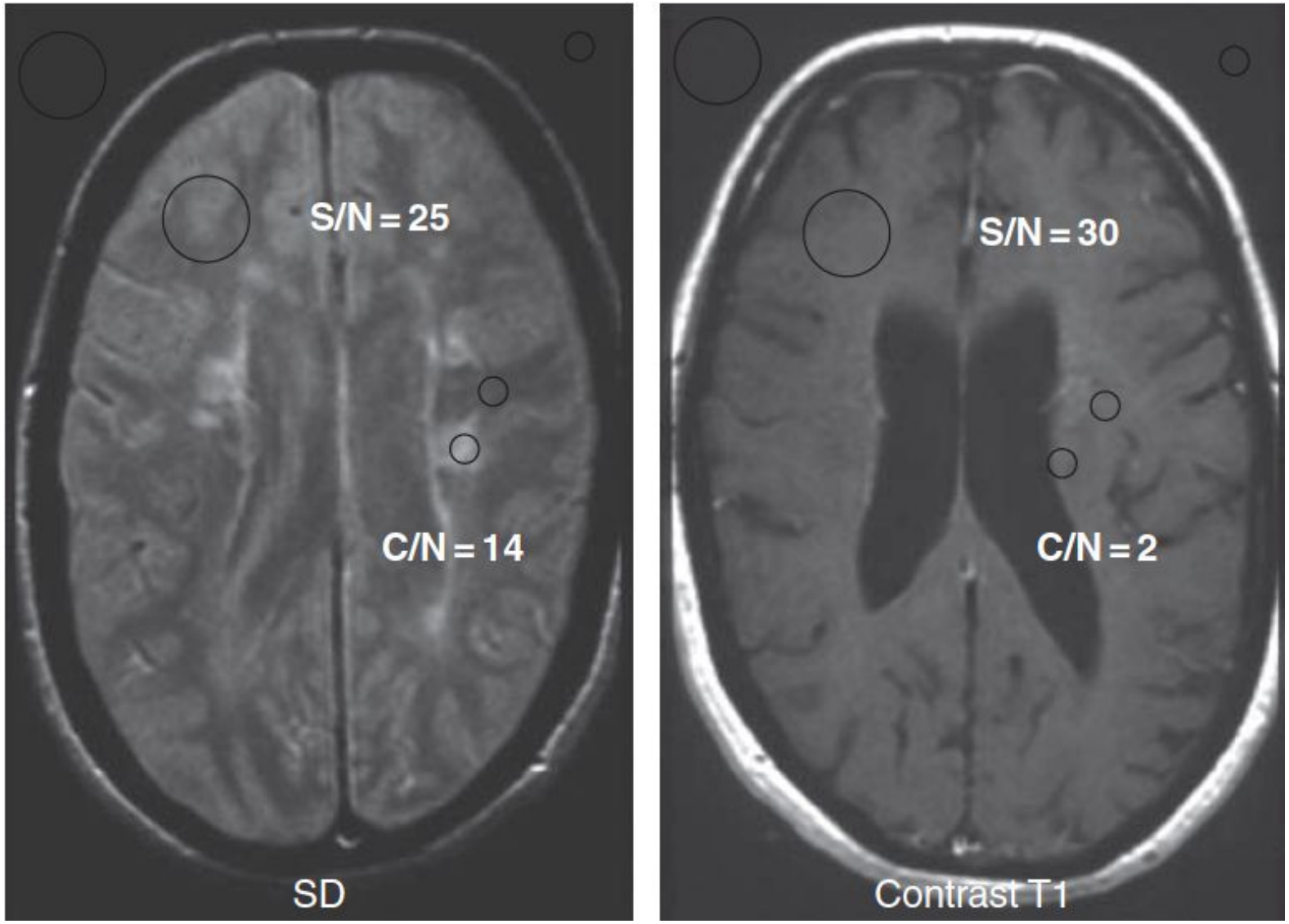
S = 700

N = 20

SNR = 700 / 20

= 35





SNR (S/N) (of normal brain) and CNR (C/N) of multiple sclerosis plaques to normal brain on spin-density and T_1 magnetic resonance images.

PART D3 · 0:45–1:00

Contrast, color, crop: the everyday trio

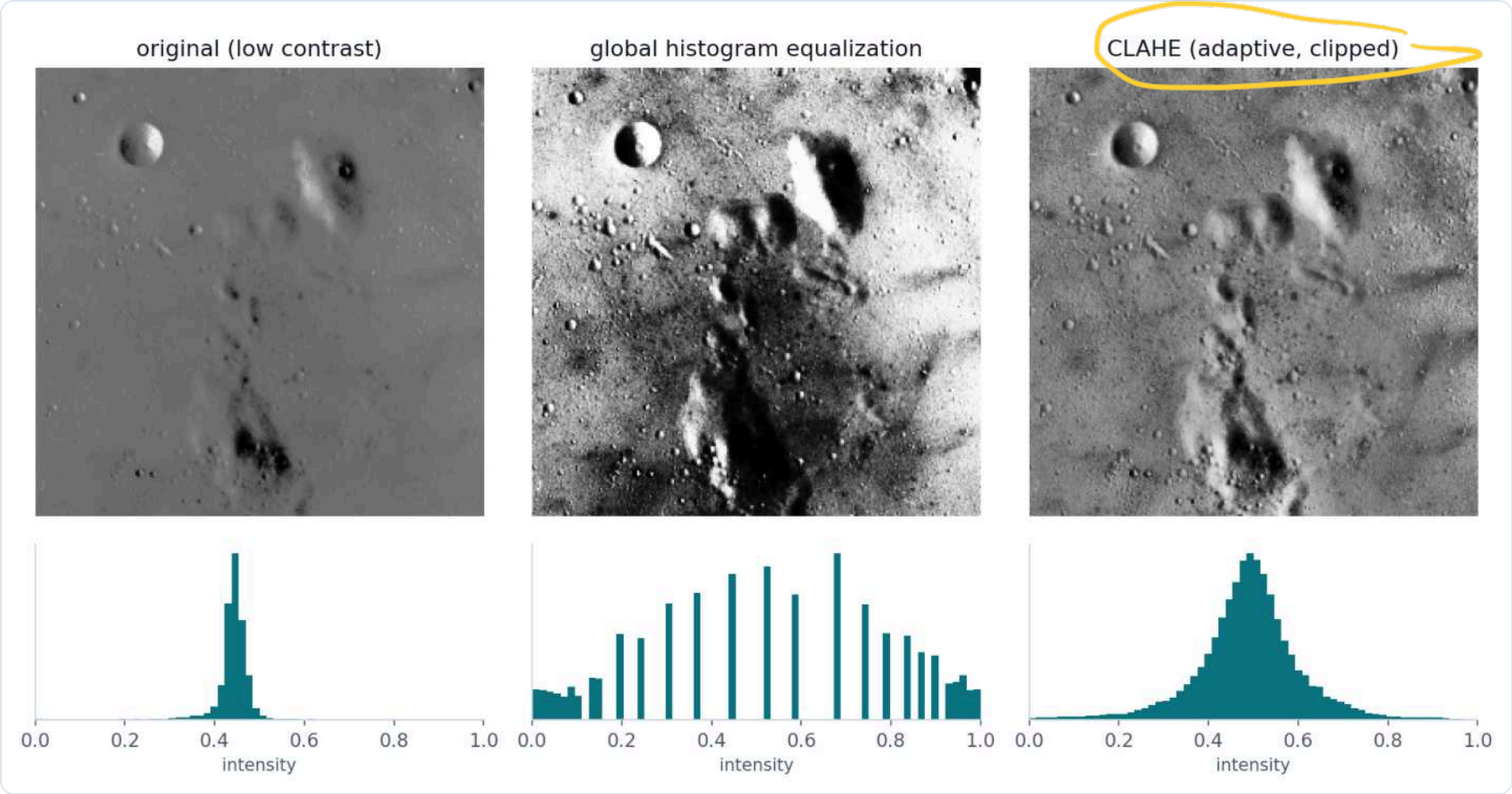
Histogram equalization and CLAHE, color-space conversions, cropping and padding. Small operations, big footprint: they appear in nearly every published pipeline.

AUTHORED

Written for this course; live examples computed with **scikit-image** and **OpenCV** on scikit-image sample images.

scikit-image.org · docs.opencv.org

Histogram equalization, global and adaptive



Global equalization spreads the histogram but can wash regions out and amplify noise; CLAHE equalizes in tiles with a clip limit. CLAHE is the standard trick on chest X-rays and fundus images.

Two lines each, one parameter that matters

```
# scikit-image
from skimage import exposure
eq    = exposure.equalize_hist(img)
clahe = exposure.equalize_adapthist(
            img, clip_limit=0.03)

# OpenCV (8-bit grayscale)
op  = cv2.createCLAHE(clipLimit=2.0,
                      tileGridSize=(8, 8))
out = op.apply(gray_u8)
```

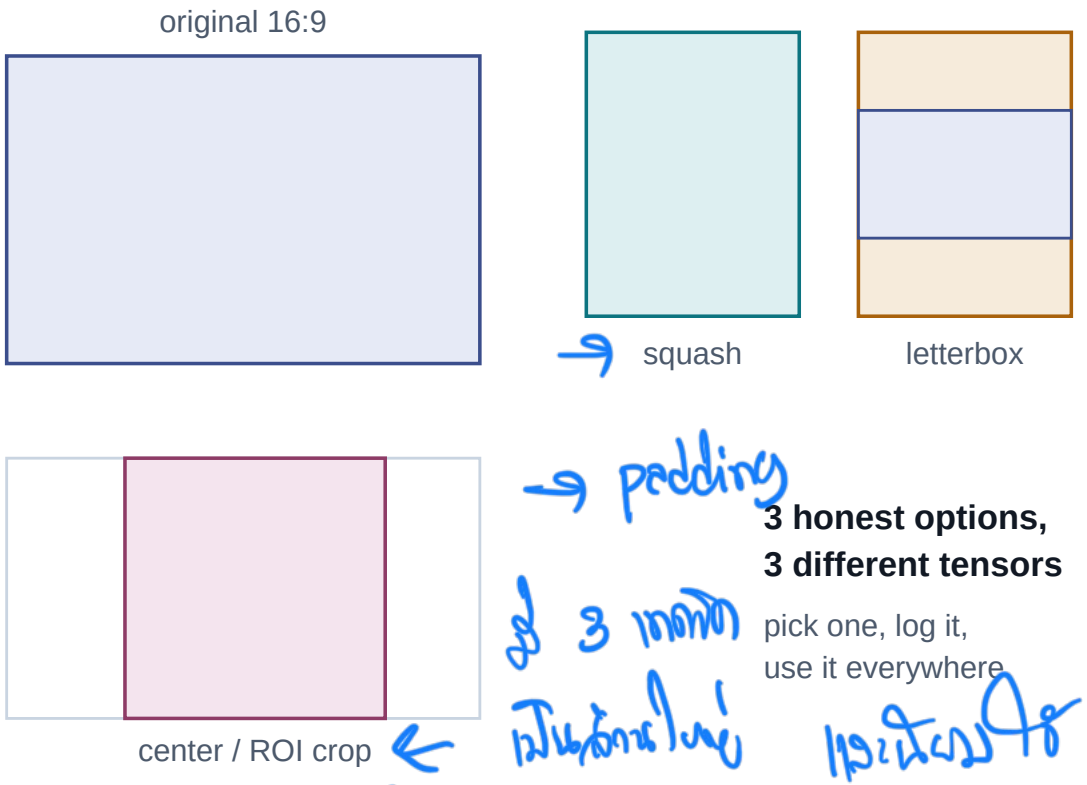
- **clip limit** controls how much contrast amplification is allowed per tile; too high and you equalize the noise.
- Report it. "CLAHE" without the clip limit and tile size is **not reproducible** (session 4 clinic, again).
- Decide **where it runs**: as offline preprocessing for every image, or as a probabilistic augmentation (Albumentations has `A.CLAHE(p=0.5)`). Those are different experiments.

Color decisions live in the pipeline, not the model

- **Gray or RGB is an experiment variable.** Dropping to one channel cuts compute and can help small ultrasound datasets; it deletes information for dermatology and histopathology. Decide with a controlled run, not by habit.
- **Be consistent end to end**: the OpenCV BGR trap from class 1 returns the moment deployment code decodes with a different library than training did.
- **Domain shift is often a color shift.** Three scanners, three color casts (recall CMU's three-camera balloon slide). Histopathology fixes it with **stain normalization** in Lab-like spaces: Reinhard, Macenko, Vahadane (course topic 3.13).
- **Ultrasound tip from our lab's work**: strip the vendor overlay and colormap before the array reaches the model; the pixels must mean tissue, not GUI.

If a human expert cannot say what an intensity means after your conversion, the model inherits the confusion.

Crop what matters, pad what remains



→ padding

3 honest options,
3 different tensors

pick one, log it,
use it everywhere

มี 3 แบบ เป็นตัวอย่าง แต่ละแบบ Crop เพราะอาจจะทำให้ทำ Data Augmentation ได้

Model ส่วนใหญ่จะต้องการ Input เป็น 4 เหลี่ยมจัตุรัส

- **Region-of-interest cropping is domain knowledge as code**: our OPG-SHAP work cropped panoramic radiographs to the mandible region before classification (course topic 3.2). Big accuracy for free, zero new labels.
- **Padding value matters**: zeros create fake black borders the model can exploit; reflect or edge padding is usually safer. Log which one you used.
- **Random crops belong to augmentation** (part F); deterministic crops belong here, applied identically at train and test.

PART E1 · 1:10–1:30

Preprocessing for the model: why optimizers care

Zero-centering, scaling, per-channel statistics. The gradient argument for why all-positive inputs slow learning, straight from the deck that modernized this material.

SOURCE DECK

Lecture 10, Training Neural Networks, Part 1, EECS 498-007 / 598-005 Deep Learning for Computer Vision, Fall 2020 · Justin Johnson, University of Michigan.
https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/ © the author · shown unmodified · selected pages

Lecture 10: Training Neural Networks (Part 1)

Consider what happens when nonlinearity is always positive

$$h_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} \sigma\left(h_j^{(\ell-1)}\right) + b_i^{(\ell)}$$

$h_i^{(\ell)}$ is the ith element of the hidden layer at layer ℓ (before activation)

$w^{(\ell)}, \mathrm{b}^{(\ell)}$ are the weights and bias of layer ℓ

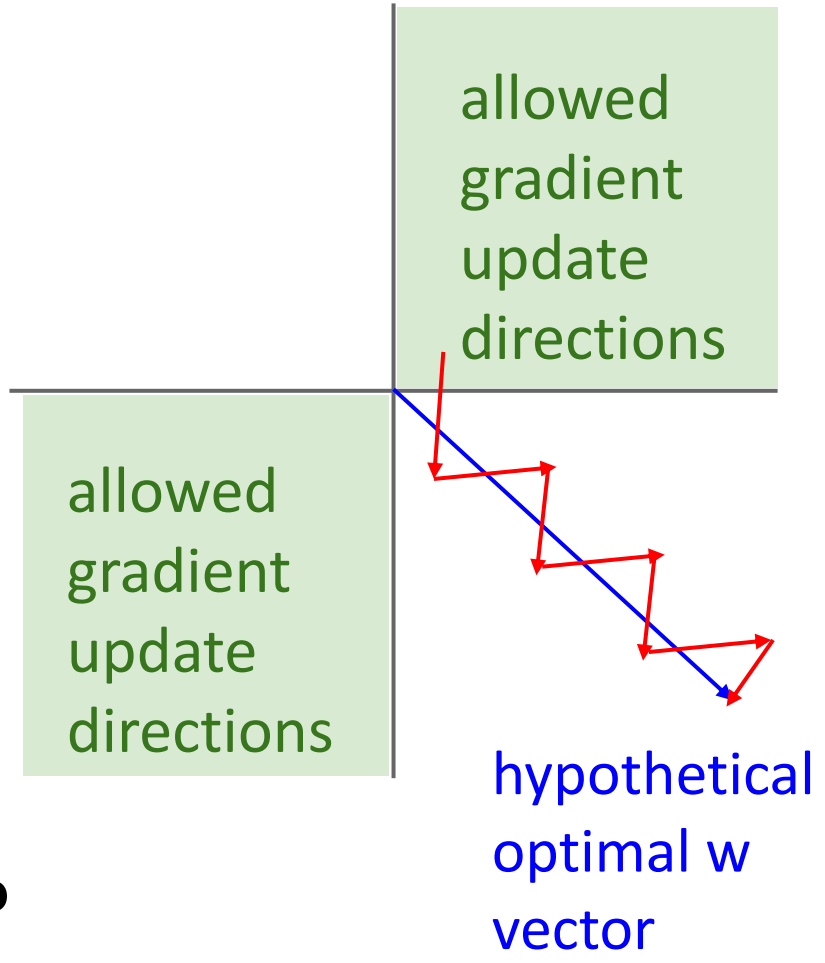
What can we say about the gradients on $w^{(\ell)}$?

Consider what happens when nonlinearity is always positive

$$h_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} \sigma\left(h_j^{(\ell-1)}\right) + b_i^{(\ell)}$$

$h_i^{(\ell)}$ is the ith element of the hidden layer at layer ℓ (before activation)

$w^{(\ell)}, \mathrm{b}^{(\ell)}$ are the weights and bias of layer ℓ

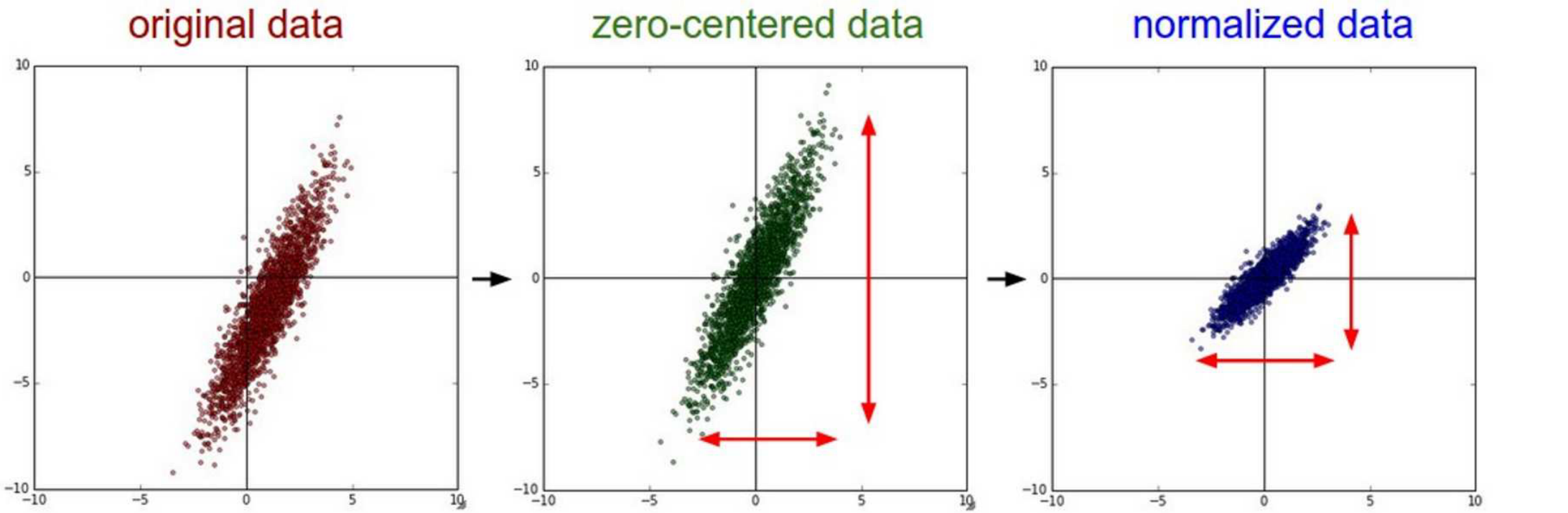


What can we say about the gradients on $w^{(\ell)}$?

Always all positive or all negative :(

Data Preprocessing

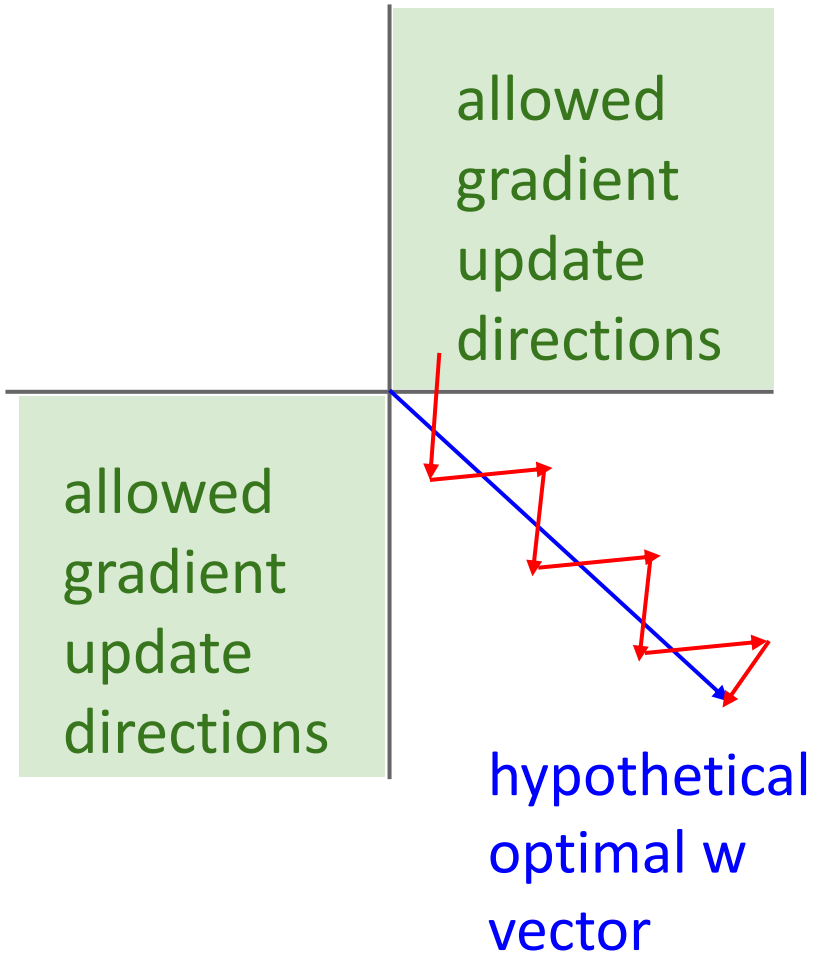
Data Preprocessing



(Assume X [NxD] is data matrix,
each example in a row)

Remember: Consider what happens when the input to a neuron is always positive...

$$h_i^{(\ell)} = \sum_j w_{i,j}^{(\ell)} \sigma\left(h_j^{(\ell-1)}\right) + b_i^{(\ell)}$$



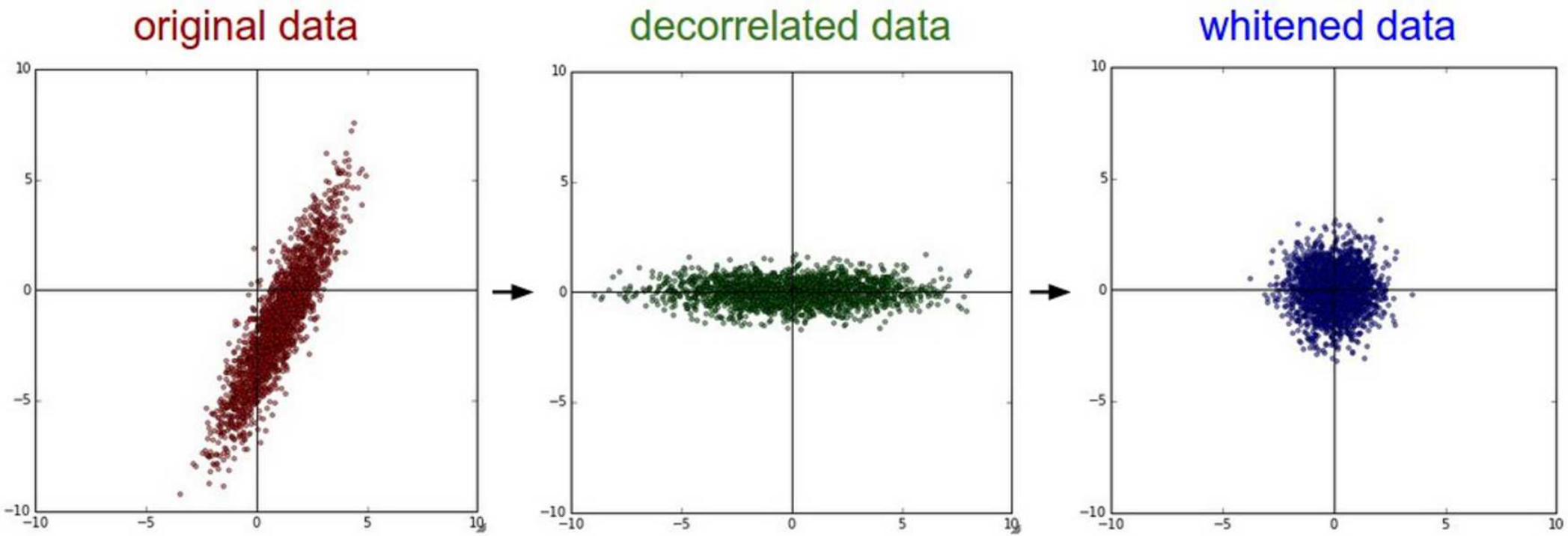
What can we say about the gradients on **w**?

Always all positive or all negative :(

(this is also why you want zero-mean data!)

Data Preprocessing

In practice, you may also see **PCA** and **Whitening** of the data

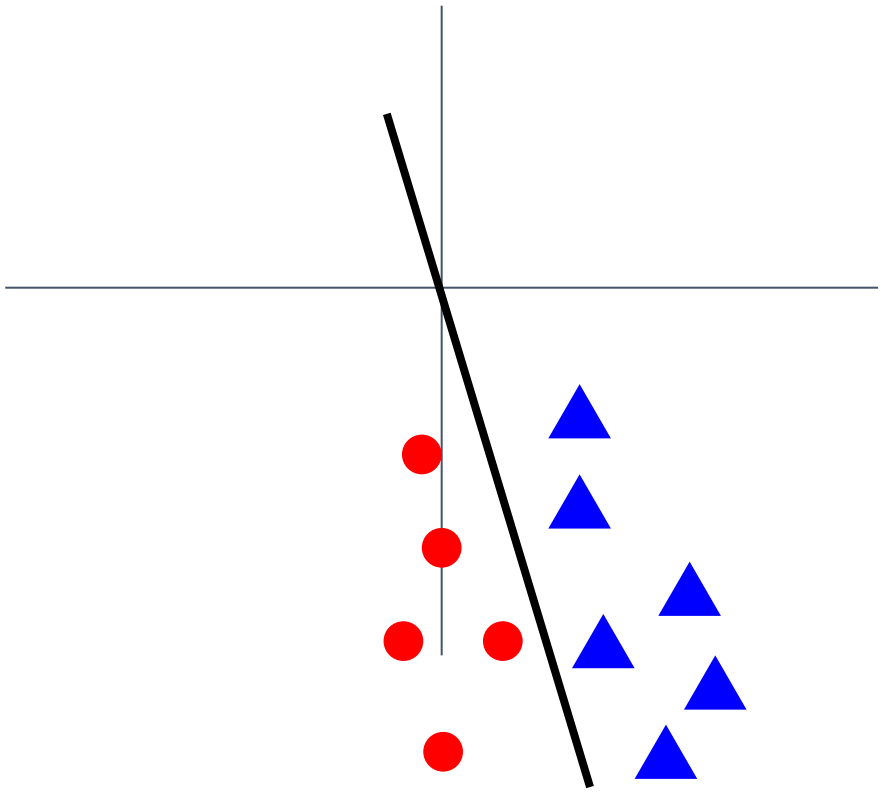


(data has diagonal covariance matrix)

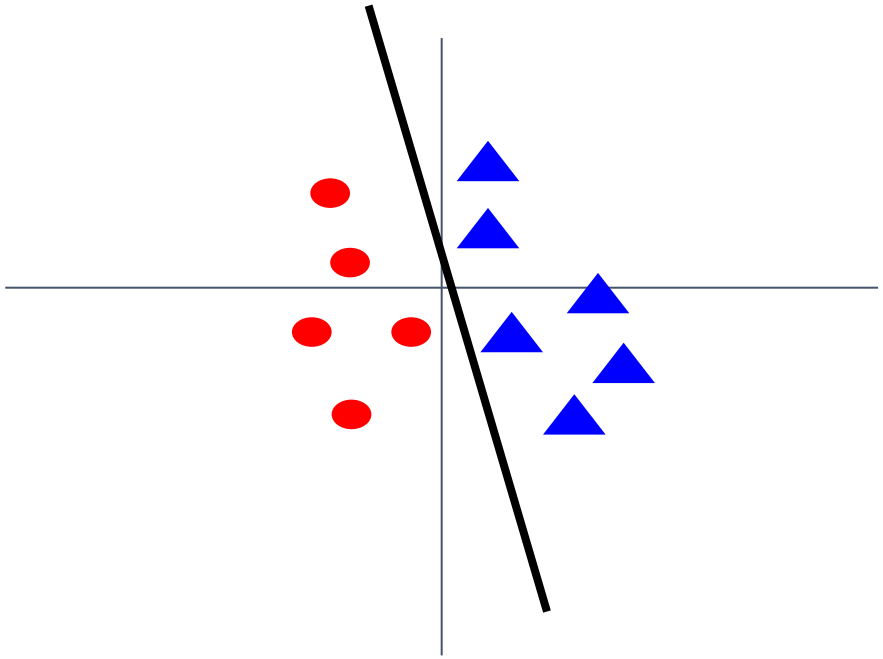
(covariance matrix is the identity matrix)

Data Preprocessing

Before normalization: classification loss very sensitive to changes in weight matrix; hard to optimize



After normalization: less sensitive to small changes in weights; easier to optimize



Data Preprocessing for Images

e.g. consider CIFAR-10 example with [32,32,3] images

- Subtract the mean image (e.g. AlexNet)
  (mean image = [32,32,3] array)
- Subtract per-channel mean (e.g. VGGNet)
  (mean along each channel = 3 numbers)
- Subtract per-channel mean and
  Divide by per-channel std (e.g. ResNet)
  (mean along each channel = 3 numbers)

Not common to do PCA or whitening

Fit preprocessing on the training split, only



Fitting on all data = the test set leaks into training.

Leakage taxonomy: Kapoor and Narayanan, Patterns 2023 (course section 1.11)

"Any preprocessing statistics (e.g. the data mean) must only be computed on the training data, and then applied to the validation / test data." (CS231n course notes, MIT license)

- Same rule at **patient level** for medical data: statistics and splits both respect the patient boundary.

The leaky version and the honest version

```
# LEAKY: statistics see the test set
mean = images.mean()          # all data!
std  = images.std()
train, test = split(images)

# HONEST: split first, fit on train only
train, test = split(images)
mean = train.mean(axis=(0,1,2))  # per channel
std  = train.std(axis=(0,1,2))
train_n = (train - mean) / std
test_n  = (test  - mean) / std   # same numbers
```

- The honest version's test transform uses **frozen numbers**; nothing about test data ever influences them.
- With a small dataset the two versions differ by little until the day the dataset shifts, and then only the honest one tells the truth.
- **nnU-Net** automates exactly this: its dataset fingerprint (spacing, intensity percentiles, class ratios) is computed from training cases and stored (course topic 3.13).

PART E2, CONTINUED · 1:30–1:50

MRI intensity standardization: normalization with anatomy in it

The same tissue lands on different intensities in every MR acquisition. Standardization learns a mapping from training images, then applies it everywhere: our rule, discovered by medical imaging decades ago.

SOURCE DECK

Image preprocessing, BME 495 Deep Learning for Biomedical Imaging, Winter 2025, Lectures 3–4 · Ulas Bagci, Northwestern University.

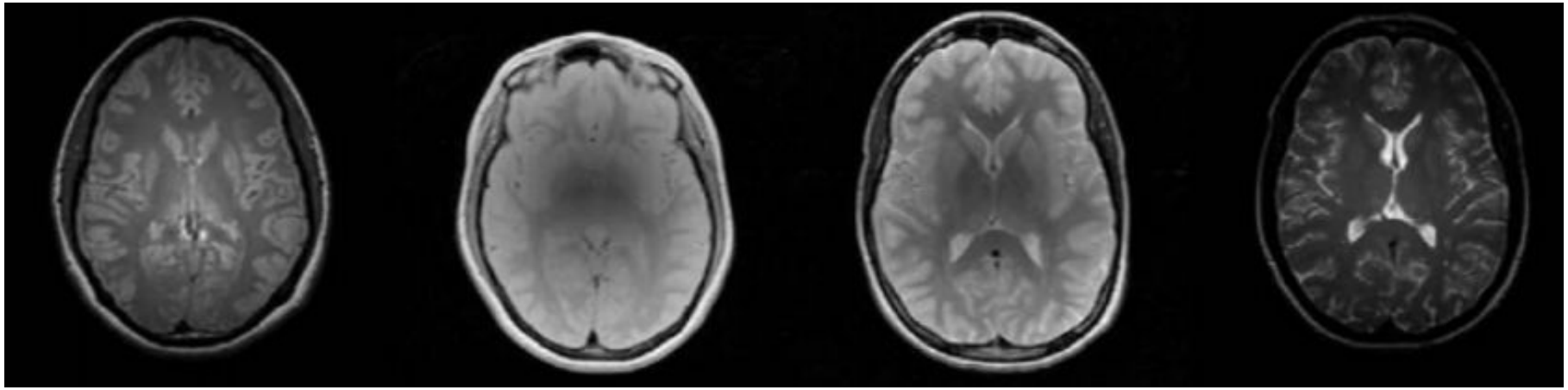
https://bagcilab.com/courses/ publicly shared by the author · shown unmodified · selected pages (method: Nyul and Udupa)

MRI Intensity Standardization

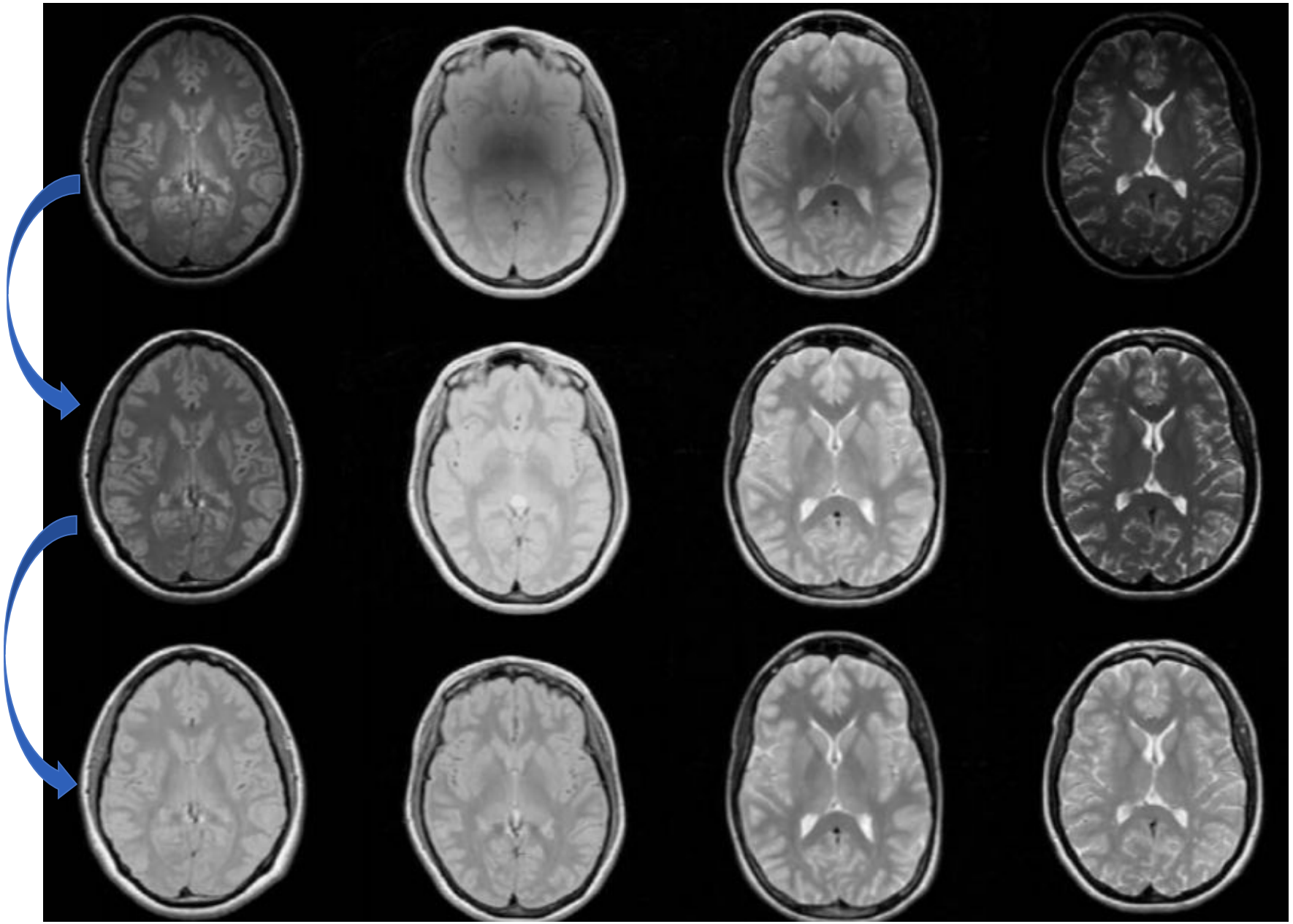


MR Intensity Non-Standardness

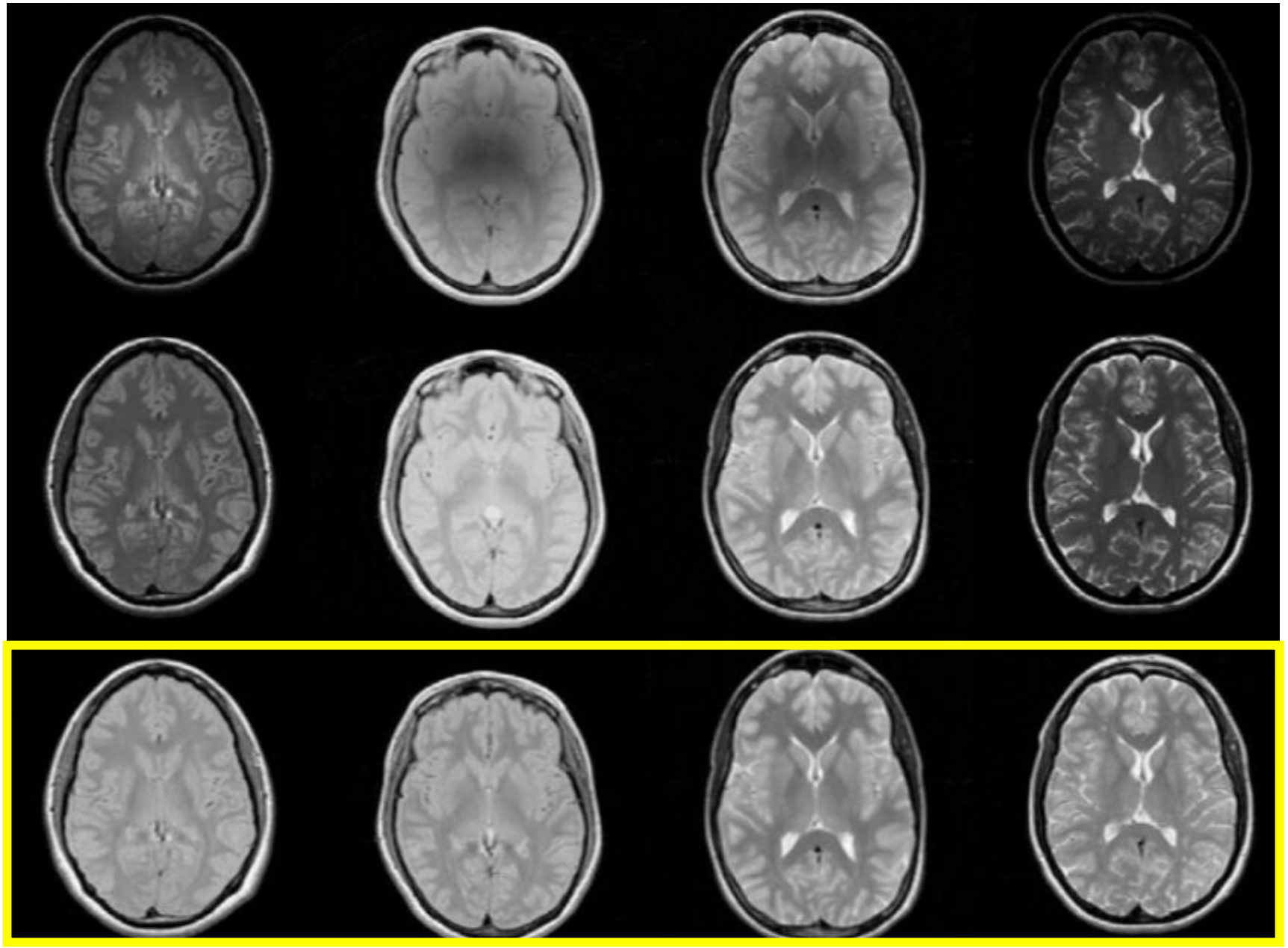
- Acquisition-to-acquisition signal intensity variations (non-standardness) are <u>inherent</u> in MR images.



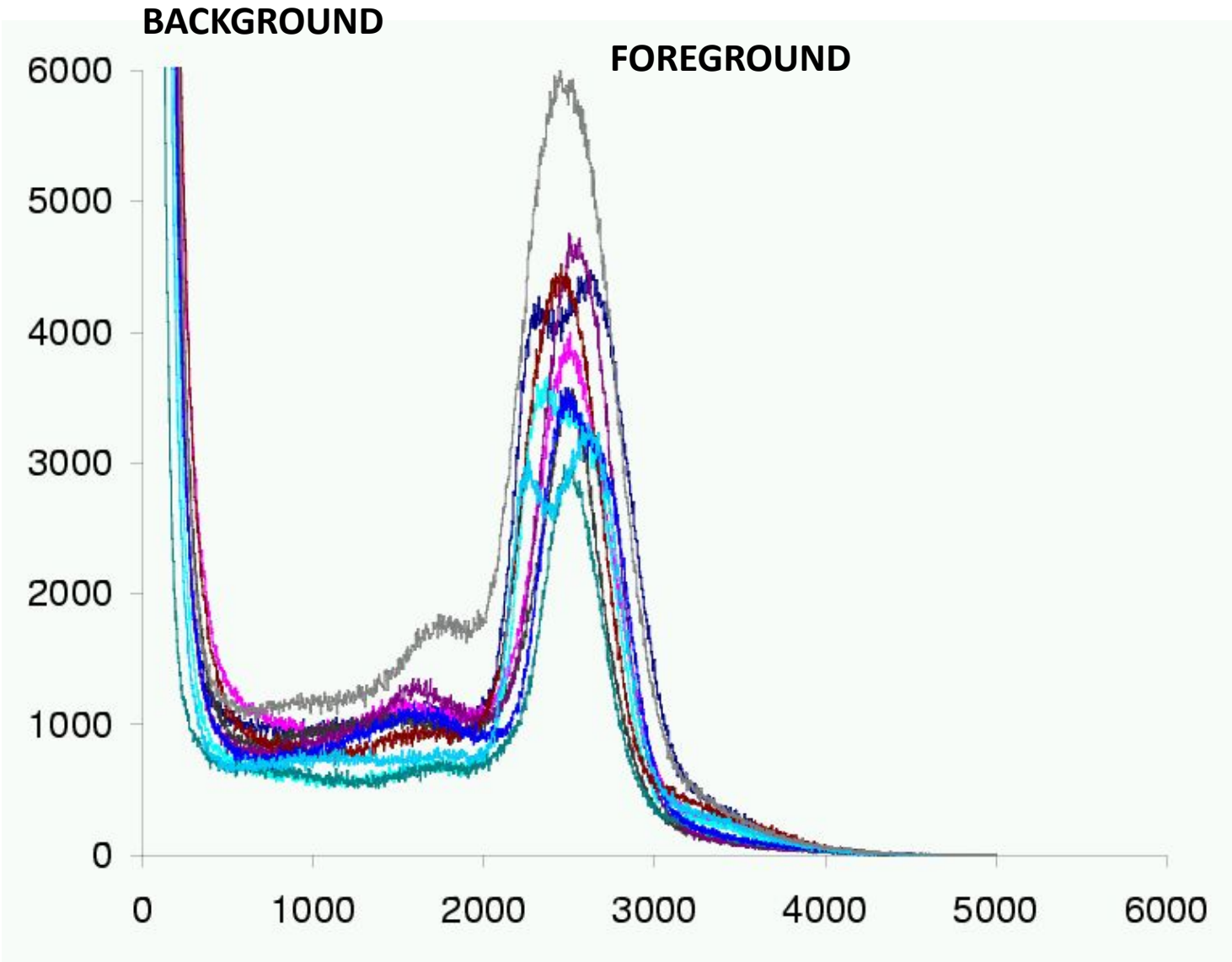
What is changed now?



Intensities are standardized!



AFTER STANDARDIZATION



10 Different PD studies

ORIGINAL HISTOGRAMS

Credit: L. Nyul

Intensity Mapping Function

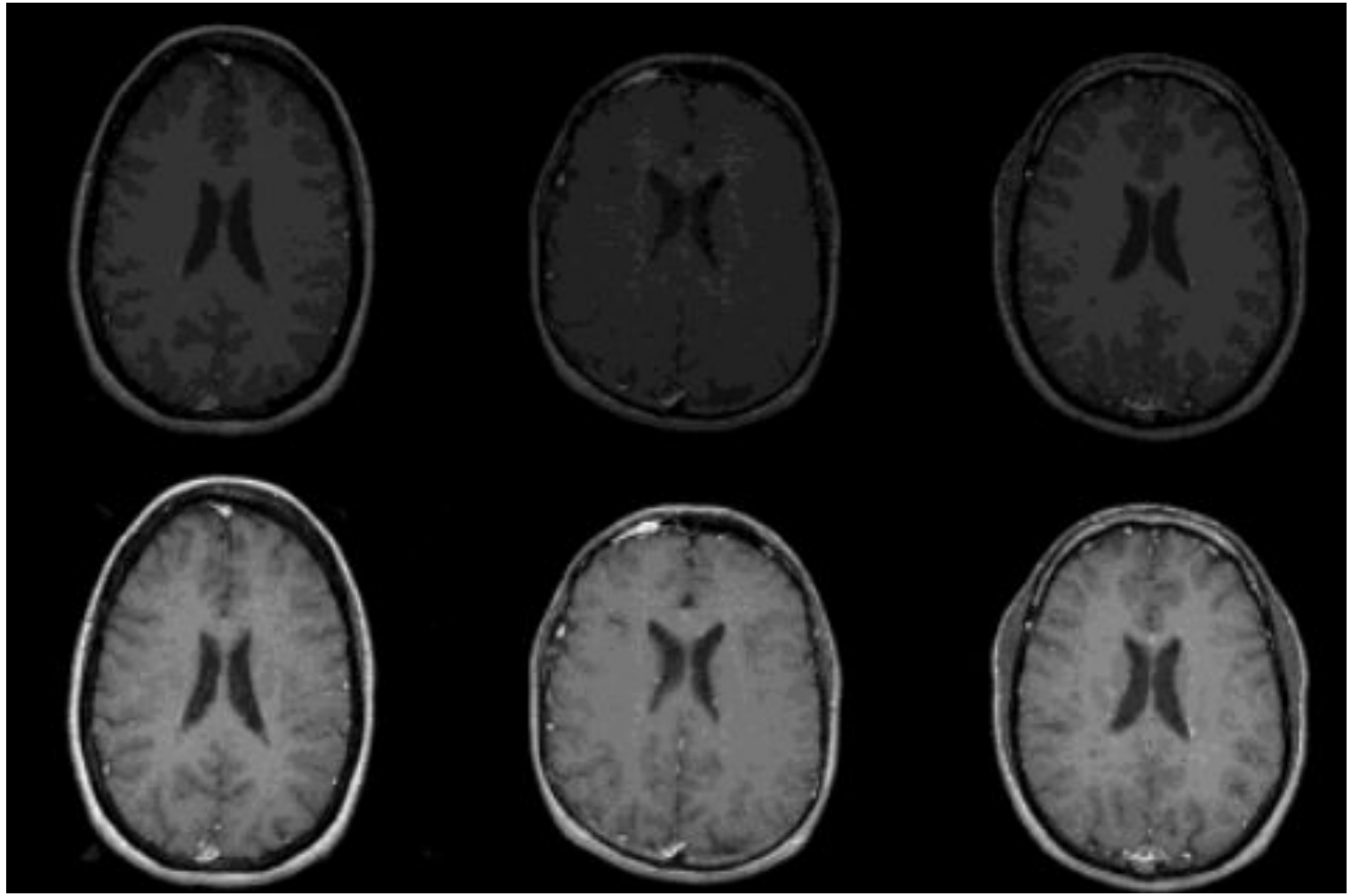
$$\tau_{\mathscr{D}_i}(x)\begin{cases}\left\lceil \mu_s + (x-\mu_i)\dfrac{s_1-\mu_s}{p_{1i}-\mu_i}\right\rceil, & \text{if } m_{1i} \le x \le \mu_i,\\ \left\lceil \mu_s + (x-\mu_i)\dfrac{s_2-\mu_s}{p_{2i}-\mu_i}\right\rceil, & \text{if } \mu_i \le x \le m_{2i},\end{cases}$$



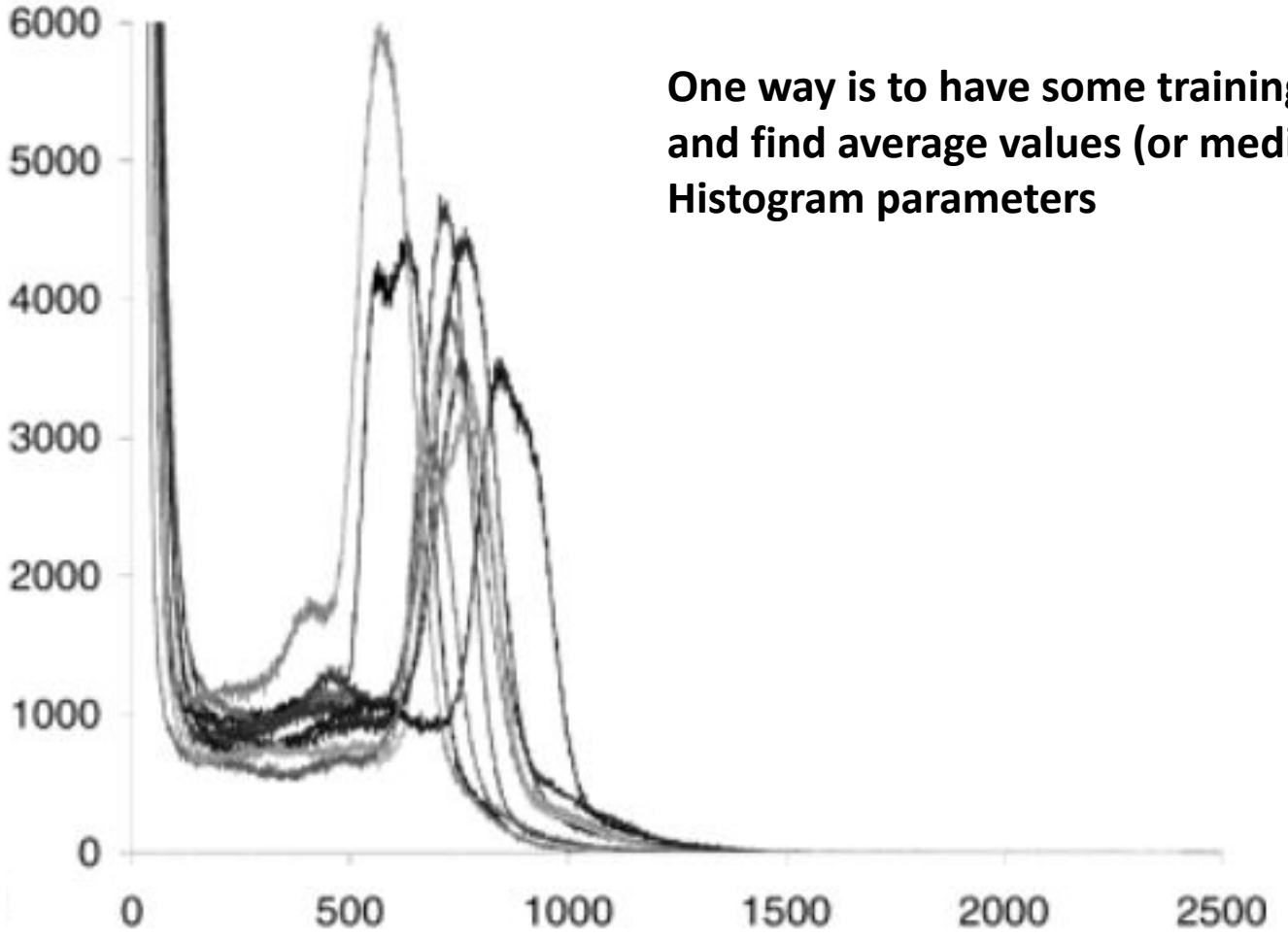
Standardized Intensity Mapping T1-MRI

Original Scale

Standard Scale

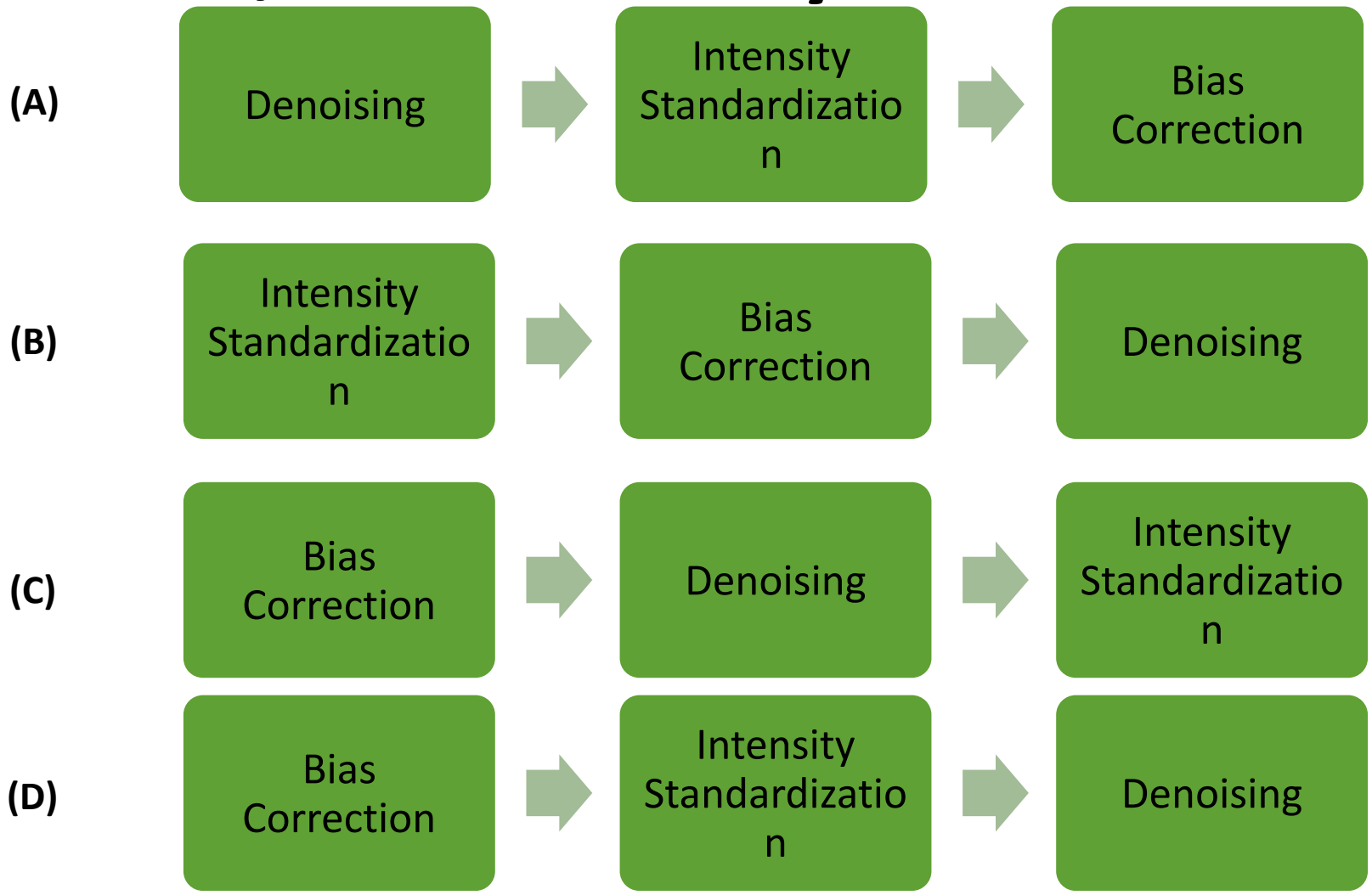


How to determine standardized parameters?



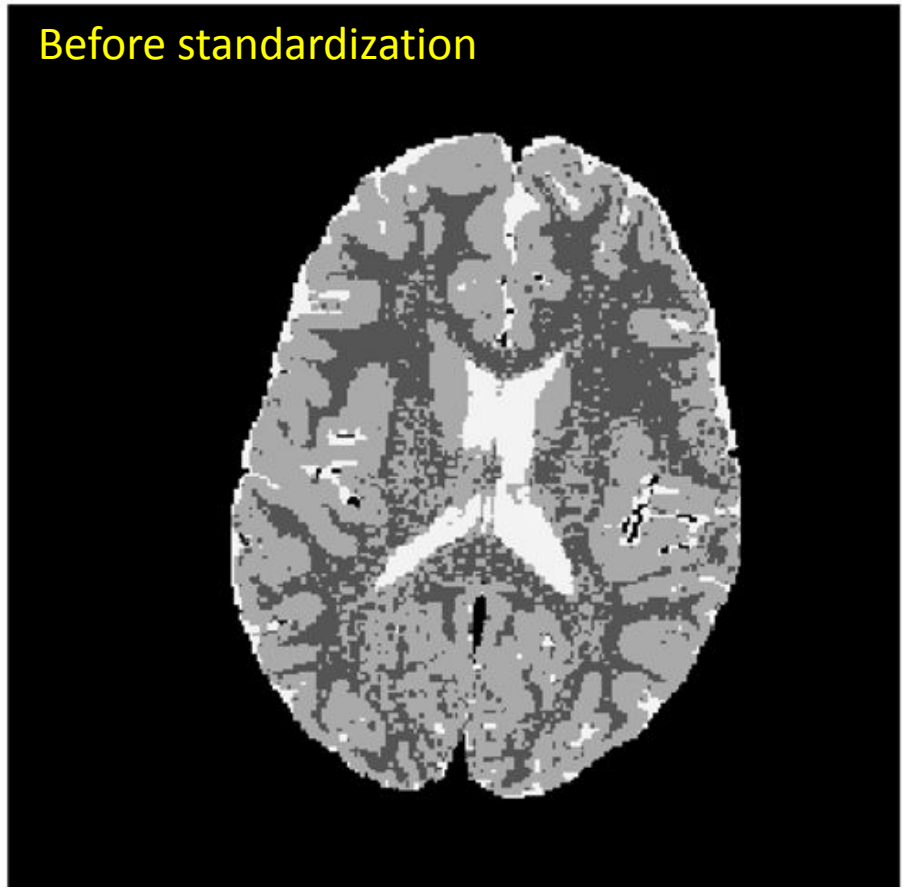
One way is to have some training images, and find average values (or median) for Histogram parameters

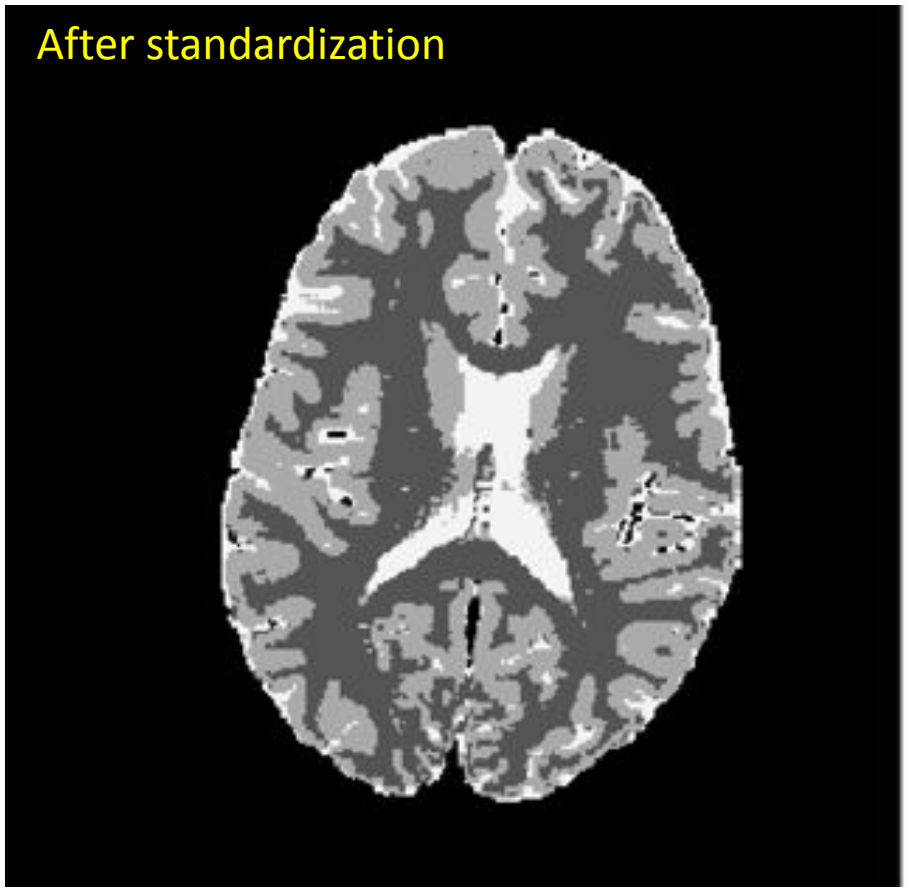
Interplay between Denoising, Bias Correction, and Intensity Standardization ?



Effects of Intensity Standardization on Image Segmentation

- Y.Zhuge et al (CVIU 2009) showed that intensity standardization simplifies brain image segmentation





Summary

- ROI, VOI, SNR, CNR, Resolution, Noise, Thresholding, Histogram…are all important terminologies to know
- Intensity non-standardness is inherent in MRI
  - must standardize
- Intensity standardization + inhomogeneity correction + denoising need to be handled prior to image analysis
- Intensity non-standardness affects
  - Perception/qualitative analysis
  - Image analysis/quantification
    - Segmentation
    - Registration
    - Recognition/localization



PART E3 · 1:50–2:05

Transfer learning: borrowed weights, borrowed statistics

Fine-tuning a pretrained backbone is the norm, and it quietly imports the preprocessing contract the backbone was trained with. Break the contract, lose the benefit.

SOURCE DECK

Lecture 7, Training Neural Networks, Part 2 (April 25, 2017), CS231n Convolutional Neural Networks for Visual Recognition · Fei-Fei Li, Justin Johnson, Serena Yeung, Stanford University.
https://cs231n.stanford.edu/ © the authors · shown unmodified · selected pages · course notes are MIT-licensed at cs231n.github.io

Lecture 7:
Training Neural Networks, Part 2

Last time: Batch Normalization

Input: $x : N \times D$

Learnable params:
$\gamma, \beta : D$

Intermediates:
$\mu, \sigma : D$
$\hat{x} : N \times D$

Output: $y : N \times D$

$$\mu_j = \frac{1}{N} \sum_{i=1}^{N} x_{i,j}$$

$$\sigma_j^2 = \frac{1}{N} \sum_{i=1}^{N} (x_{i,j} - \mu_j)^2$$

$$\hat{x}_{i,j} = \frac{x_{i,j} - \mu_j}{\sqrt{\sigma_j^2 + \varepsilon}}$$

$$y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j$$

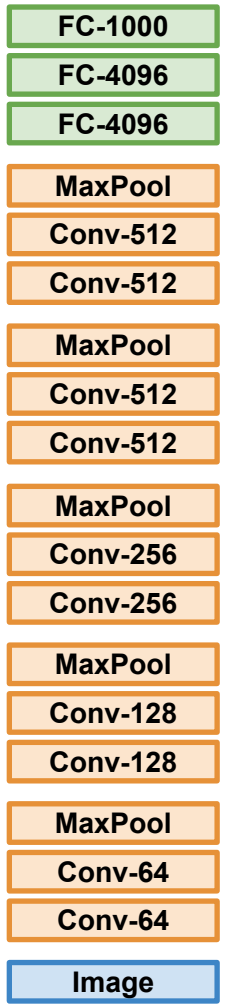
Transfer Learning

"You need a lot of a data if you want to train/use CNNs"

Transfer Learning

"You need a lot of a data if you want to train/use CNNs"

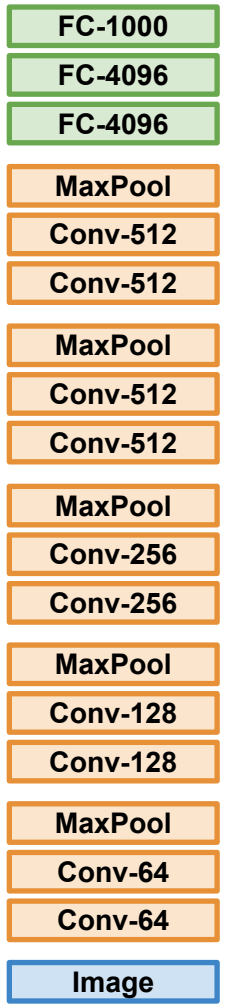
BUSTED



More specific

More generic

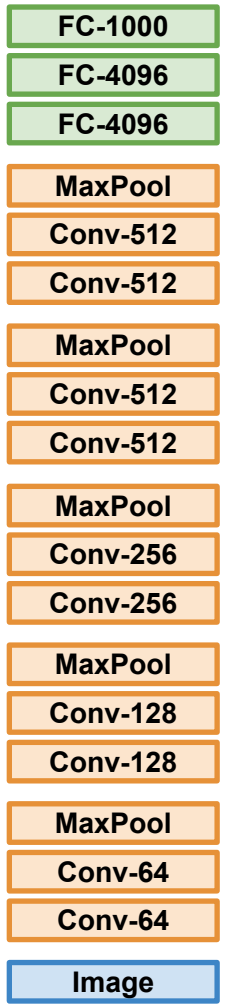
	very similar dataset	very different dataset
very little data	?	?
quite a lot of data	?	?



More specific

More generic

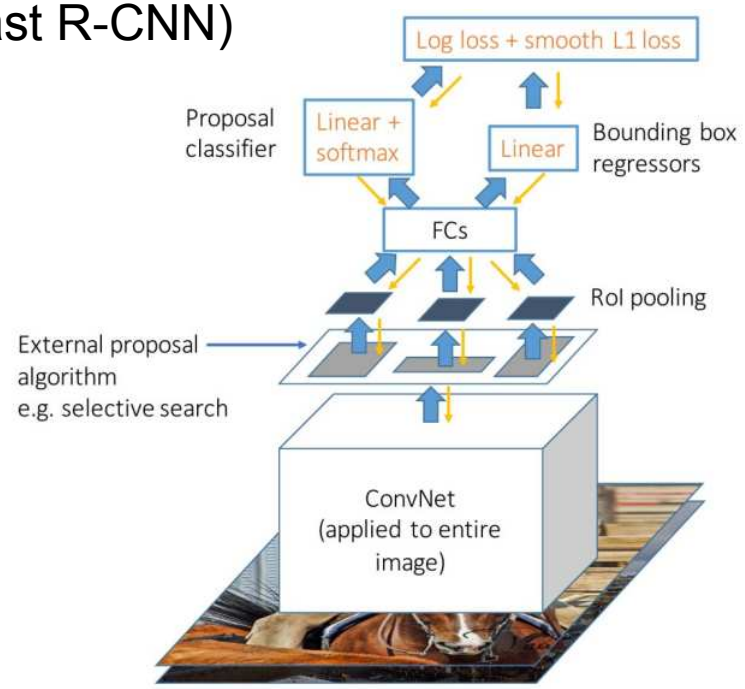
	very similar dataset	**very different dataset**
very little data	Use Linear Classifier on top layer	?
quite a lot of data	Finetune a few layers	?



	very similar dataset	very different dataset
very little data	Use Linear Classifier on top layer	You're in trouble… Try linear classifier from different stages
quite a lot of data	Finetune a few layers	Finetune a larger number of layers

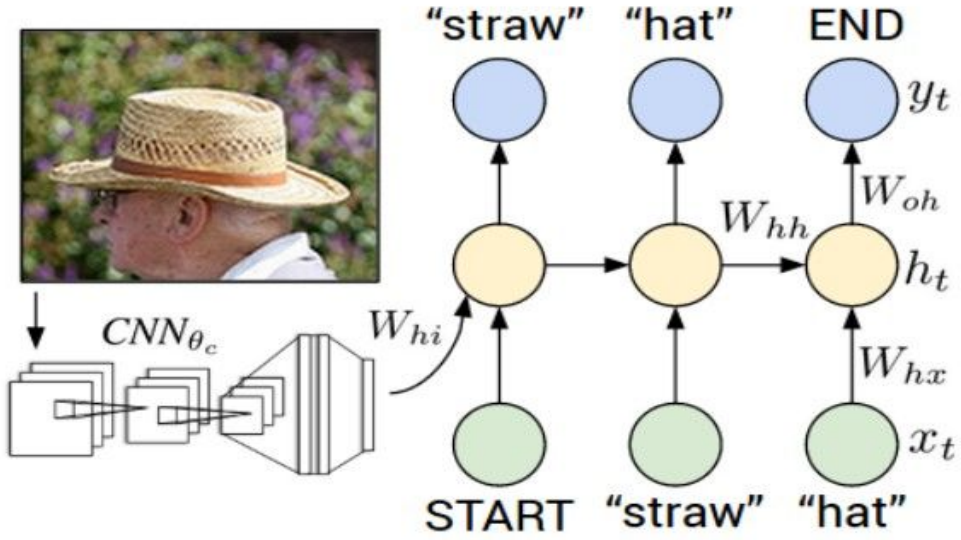
Transfer learning with CNNs is pervasive…
(it's the norm, not an exception)

Object Detection
(Fast R-CNN)



Girshick, "Fast R-CNN", ICCV 2015
Figure copyright Ross Girshick, 2015. Reproduced with permission.

Image Captioning: CNN + RNN



Karpathy and Fei-Fei, "Deep Visual-Semantic Alignments for Generating Image Descriptions", CVPR 2015
Figure copyright IEEE, 2015. Reproduced for educational purposes.

Takeaway for your projects and beyond:

Have some dataset of interest but it has < ~1M images?

1. Find a very large dataset that has similar data, train a big ConvNet there
2. Transfer learn to your dataset

Deep learning frameworks provide a “Model Zoo” of pretrained models so you don’t need to train your own

Caffe: https://github.com/BVLC/caffe/wiki/Model-Zoo
TensorFlow: https://github.com/tensorflow/models
PyTorch: https://github.com/pytorch/vision

Those six numbers you keep copy-pasting

```python
# The ImageNet statistics behind most backbones
mean = [0.485, 0.456, 0.406]
std  = [0.229, 0.224, 0.225]

tfm = v2.Compose([
    v2.Resize(256, antialias=True),
    v2.CenterCrop(224),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize(mean=mean, std=std),
])
```

torchvision v2; identical idea in every framework.

- They are the **training-set statistics of ImageNet**: the rule from E2, applied by someone else, frozen into the checkpoint's contract.
- **Match the backbone**: many ViT and CLIP checkpoints use different constants (0.5-based, or CLIP's own). Read the model card, do not assume.
- Grayscale medical images through an RGB backbone: replicate to 3 channels **then** use the backbone's stats, or retrain the stem; document which.
- Wrong stats will not crash. They just shift every input off the distribution the network expects.

PART F1 · 2:05–2:25

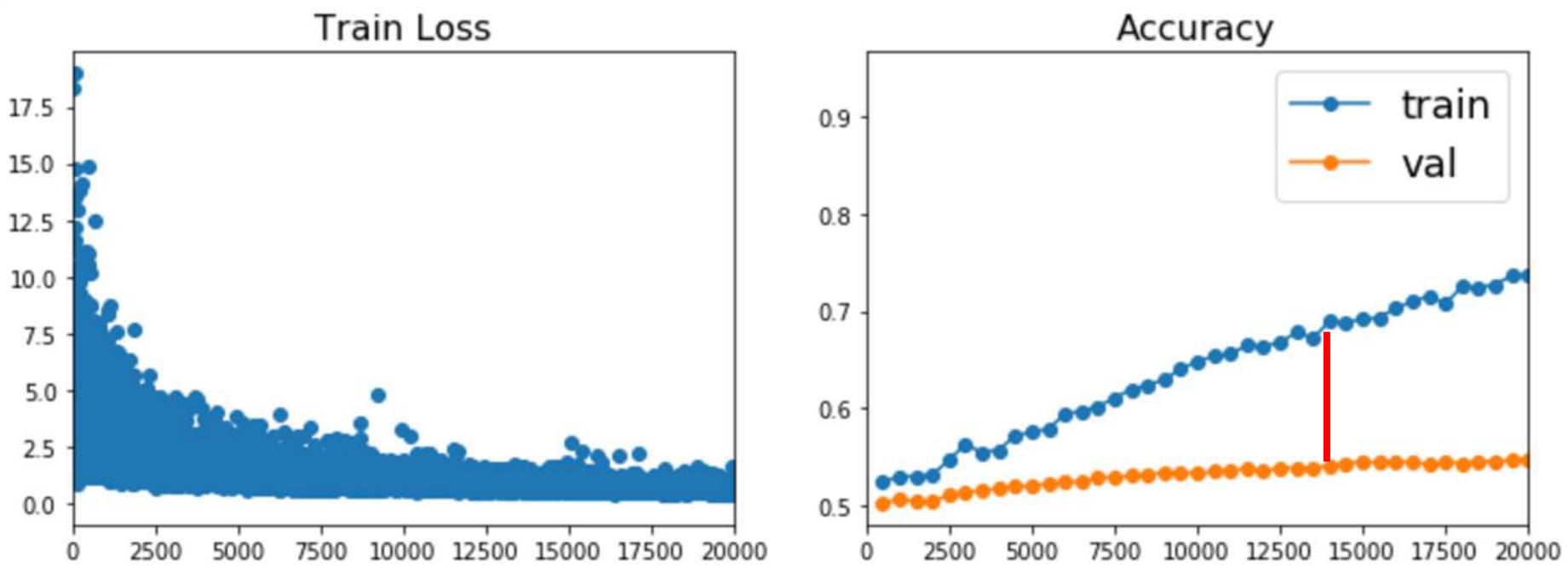
Augmentation: preprocessing that changes every epoch

Train-time randomness, test-time averaging. Flips, crops, color jitter, Cutout, MixUp: one pattern, many instances, and a knob for how much dataset you can fake.

SOURCE DECK

Lecture 10, Training Neural Networks, Part 1, EECS 498-007 / 598-005, Fall 2020 · Justin Johnson, University of Michigan.
https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/ © the author · shown unmodified · selected pages

Now your model is training … but it overfits!



Regularization

Regularization: A common pattern

Training: Add some kind of randomness

$$y = f_W(x, z)$$

Testing: Average out randomness (sometimes approximate)

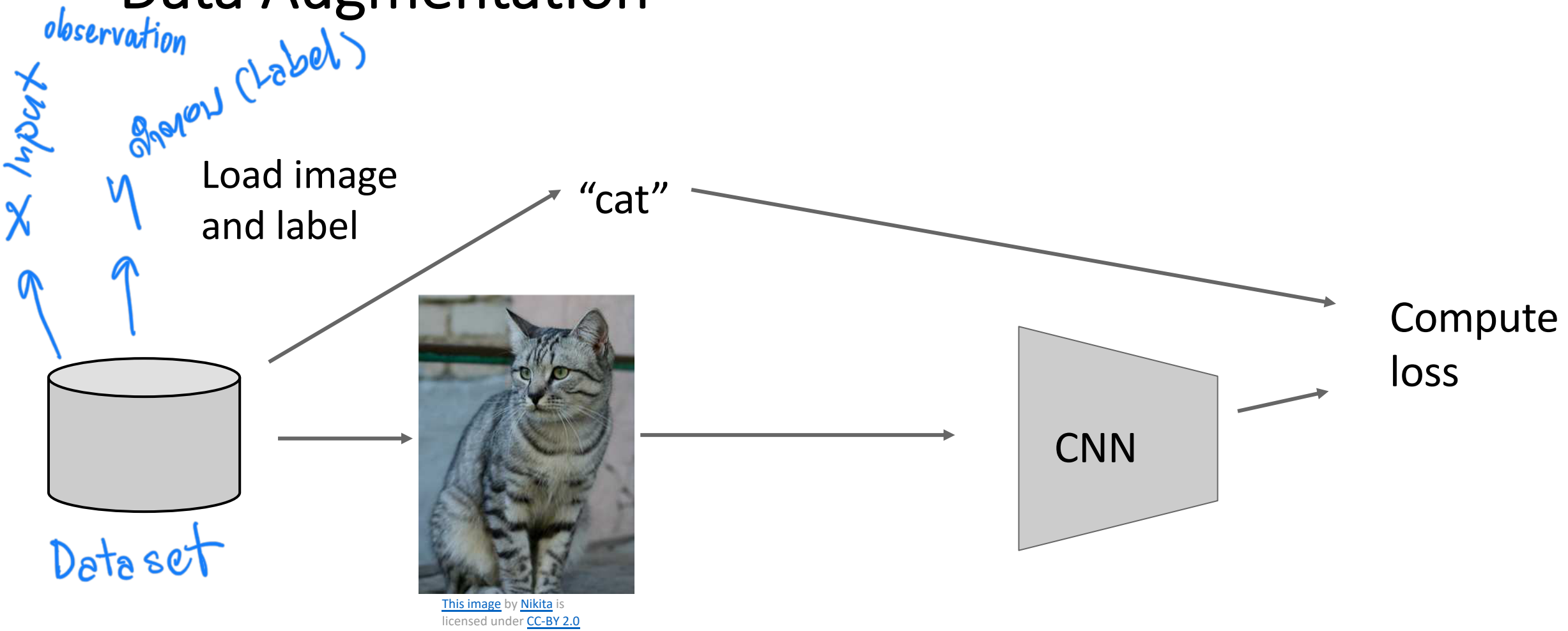
$$y = f(x) = E_z[f(x, z)] = \int p(z) f(x, z) dz$$

Example: Batch Normalization

Training: Normalize using stats from random minibatches

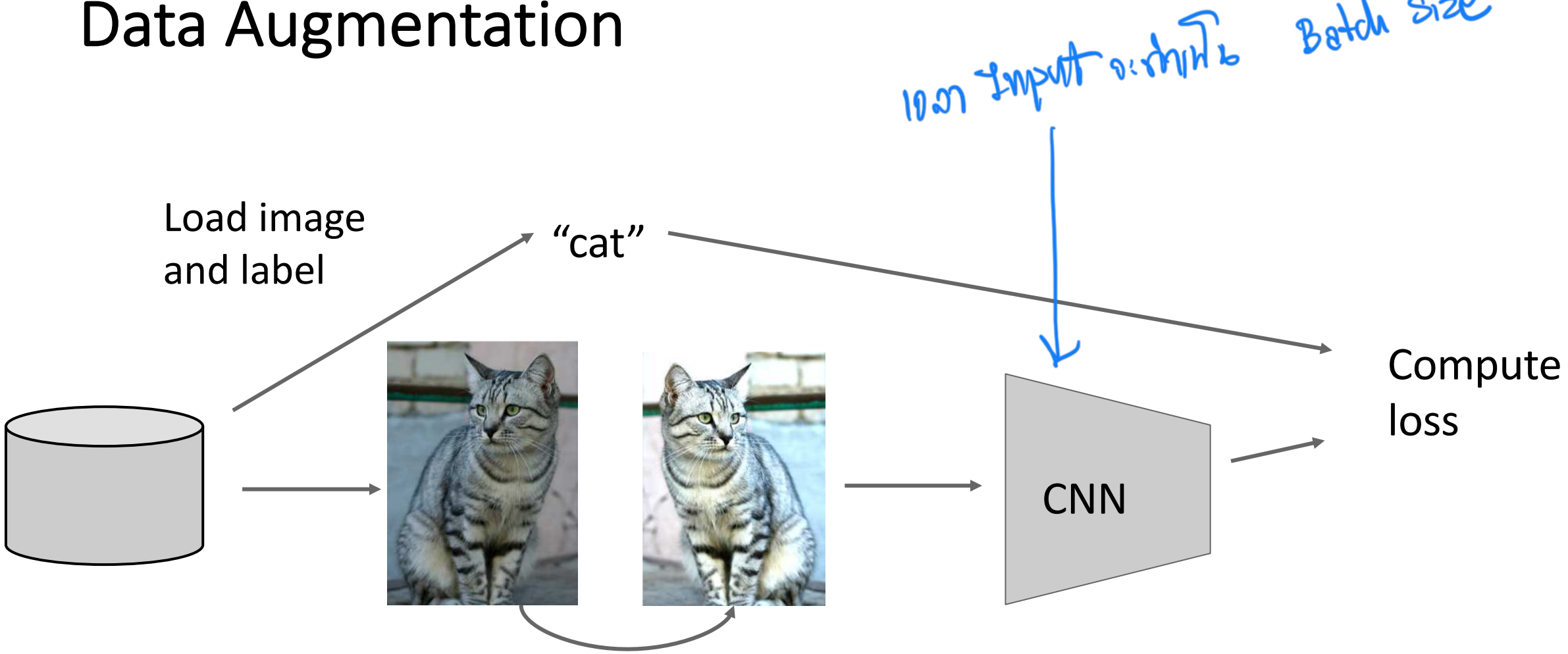
Testing: Use fixed stats to normalize

Data Augmentation



This image by Nikita is licensed under CC-BY 2.0

Data Augmentation



Data Augmentation: Horizontal Flips



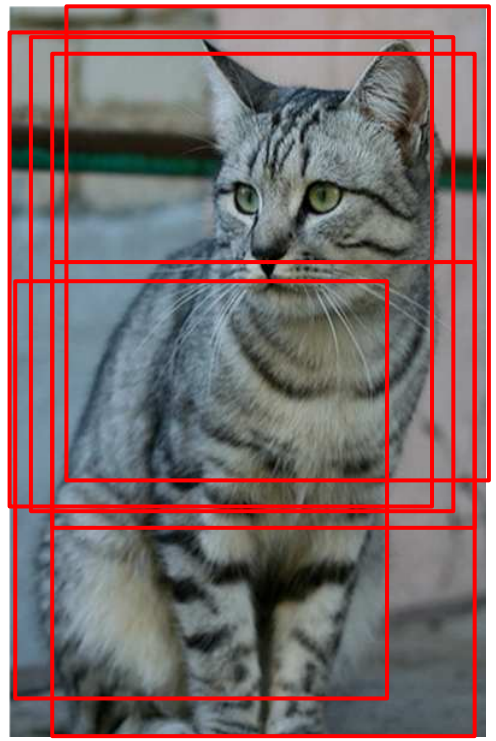


Data Augmentation: Random Crops and Scales

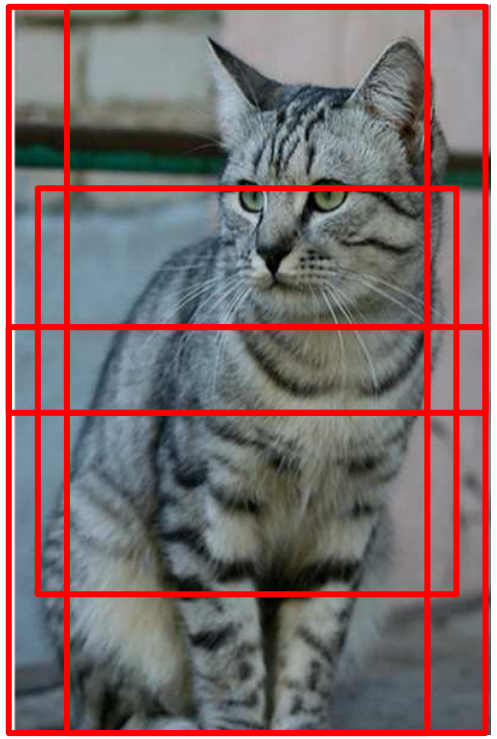
Training: sample random crops / scales

ResNet:

1. Pick random L in range [256, 480]
2. Resize training image, short side = L
3. Sample random 224 x 224 patch



Data Augmentation: Random Crops and Scales



Training: sample random crops / scales

ResNet:

1. Pick random L in range [256, 480]
2. Resize training image, short side = L
3. Sample random 224 x 224 patch

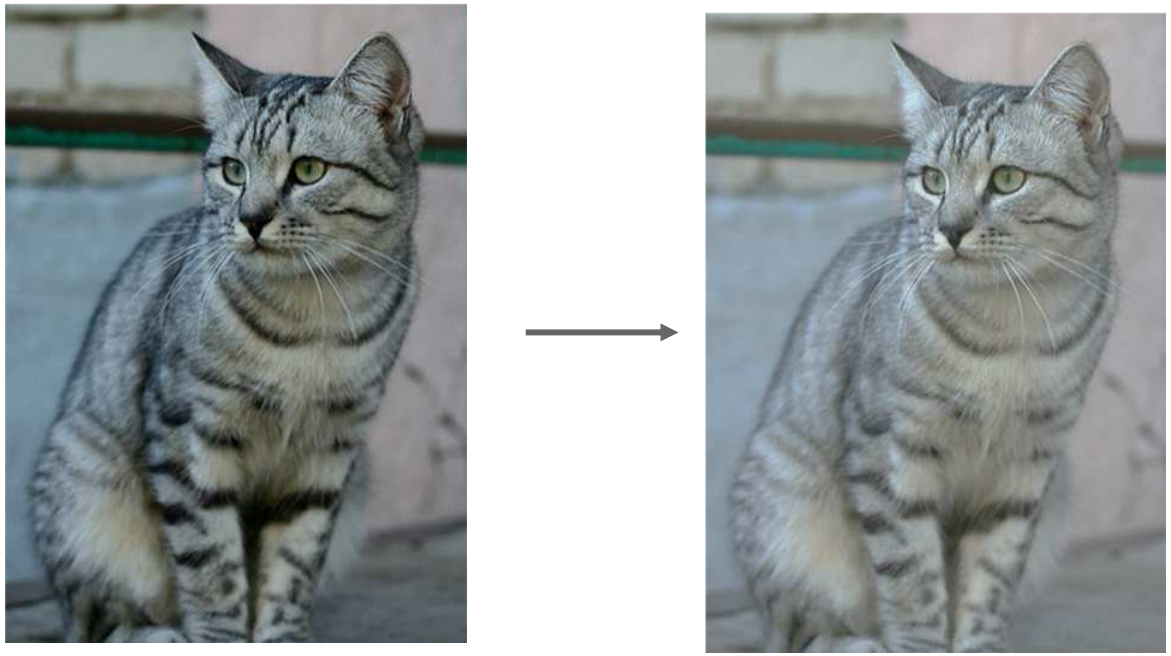
Testing: average a fixed set of crops

ResNet:

1. Resize image at 5 scales: {224, 256, 384, 480, 640}
2. For each size, use 10 224 x 224 crops: 4 corners + center, + flips

Data Augmentation: Color Jitter

Simple: Randomize contrast and brightness



More Complex:

1. Apply PCA to all [R, G, B] pixels in training set
2. Sample a "color offset" along principal component directions
3. Add offset to all pixels of a training image

(Used in AlexNet, ResNet, etc)

Data Augmentation: Get creative for your problem!

Random mix/combinations of :

- translation
- rotation
- stretching
- shearing,
- lens distortions, … (go crazy)

Regularization: A common pattern

Training: Add some randomness

Testing: Marginalize over randomness

Examples:

Dropout

Batch Normalization

Data Augmentation

Wan et al, “Regularization of Neural Networks using DropConnect”, ICML 2013

Regularization: Stochastic Depth

Training: Set random images regions to 0

Testing: Use the whole image

Examples:

Dropout

Batch Normalization

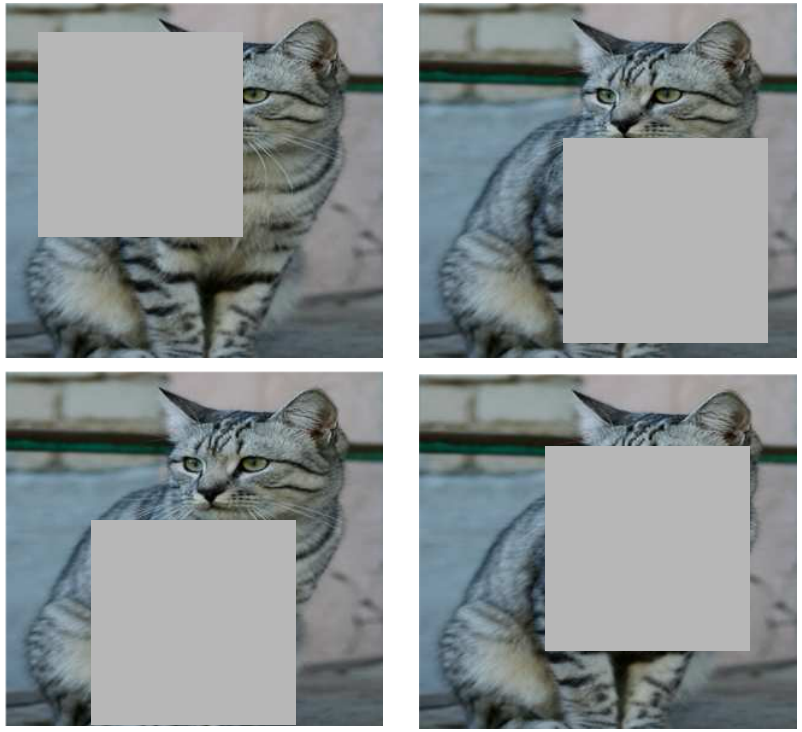
Data Augmentation

DropConnect

Fractional Max Pooling

Stochastic Depth

Cutout



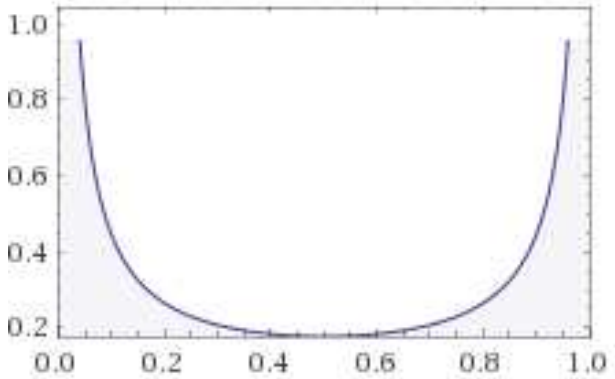
Works very well for small datasets like CIFAR, less common for large datasets like ImageNet

DeVries and Taylor, "Improved Regularization of Convolutional Neural Networks with Cutout", arXiv 2017

Regularization: Mixup

Training: Train on random blends of images

Testing: Use original images



Sample blend probability from a beta distribution Beta(a, b) with a=b≈0 so blend weights are close to 0/1

Examples:

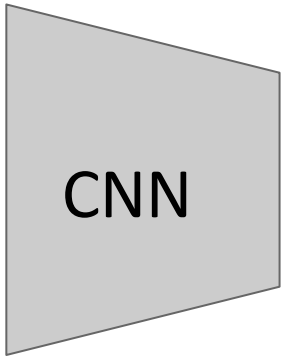
Dropout
Batch Normalization
Data Augmentation
DropConnect
Fractional Max Pooling
Stochastic Depth
Cutout
Mixup







Randomly blend the pixels of pairs of training images, e.g. 40% cat, 60% dog



Target label:
cat: 0.4
dog: 0.6

ตัวไหน?

How ?

ก็ Train ไหล่

Zhang et al, "*mixup*: Beyond Empirical Risk Minimization", ICLR 2018

Regularization: Mixup

Training: Train on random blends of images
Testing: Use original images

Examples:

Dropout
Batch Normalization
Data Augmentation
DropConnect
Fractional Max Pooling
Stochastic Depth
Cutout
Mixup

- Consider dropout for large fully-connected layers
- Batch normalization and data augmentation almost always a good idea
- Try cutout and mixup especially for small classification datasets

Zhang et al, "*mixup*: Beyond Empirical Risk Minimization", ICLR 2018

Summary

1. **One time setup**
   Activation functions, data preprocessing, weight initialization, regularization

Today

2. **Training dynamics**
   Learning rate schedules; large-batch training; hyperparameter optimization
3. **After training**
   Model ensembles, transfer learning

Next time

PART F2 · 2:25–2:40

Medical augmentation: fake data, real physics

Ghosting, spike noise, motion, bias fields: medical augmentation can simulate the scanner itself, not just geometry. Plus the honest limits: where MixUp should not go.

SOURCE DECK

Lecture 9, Evaluating DL Algorithms and Data Augmentation, BME 495, Winter 2025 · Ulas Bagci, Northwestern University.

https://bagcilab.com/courses/ publicly shared by the author · shown unmodified · selected pages (review: Chlap et al. 2021)

Data Augmentation

a technique used in machine learning and deep learning to increase the diversity of data available for training models, without actually collecting new data.

- **Common Techniques:** Rotation, flipping, cropping, noise addition, affine transformations.

- **Advanced Techniques:** Generative models (e.g., GANs), MixUp, etc.

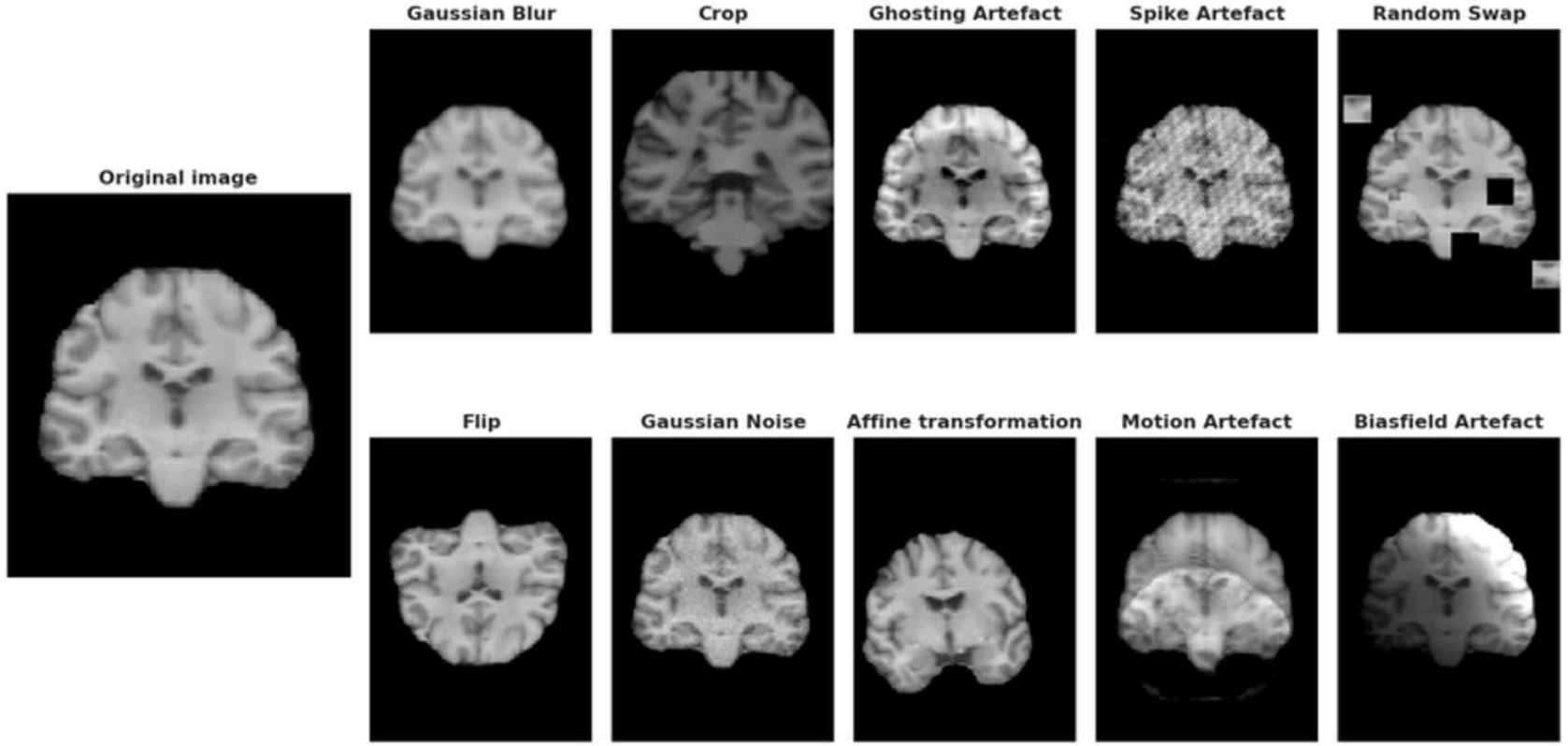
Data Augmentation

a technique used in machine learning and deep learning to increase the diversity of data available for training models, without actually collecting new data.

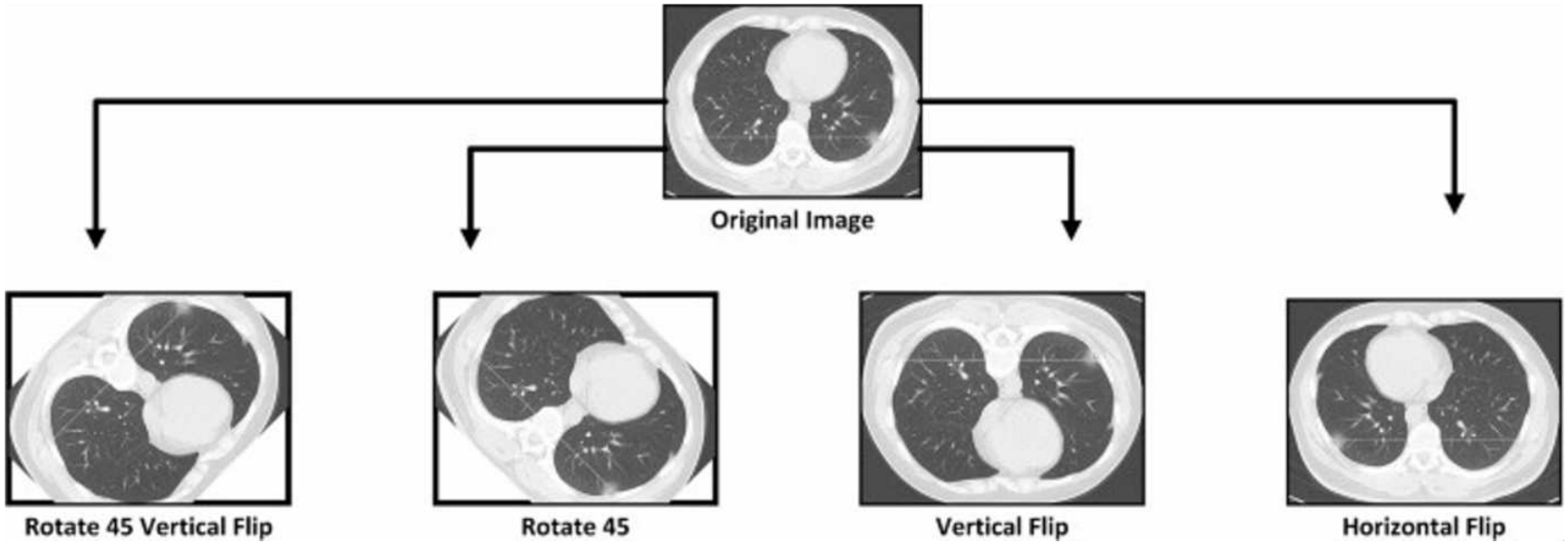
Benefits of Data Augmentation:

- **Improved Model Performance**: Augmentation can lead to better generalization, reducing the risk of overfitting, especially in deep learning models that require large amounts of training data.
- **Handling Imbalanced Datasets**: It can help balance datasets by augmenting underrepresented classes.
- **Enhancing Robustness**: Models trained with augmented data can handle real-world variability (like different lighting conditions in images) more effectively.
- **Cost-Effective**: Augmentation provides a way to artificially expand datasets without the need and expense of actual data collection.

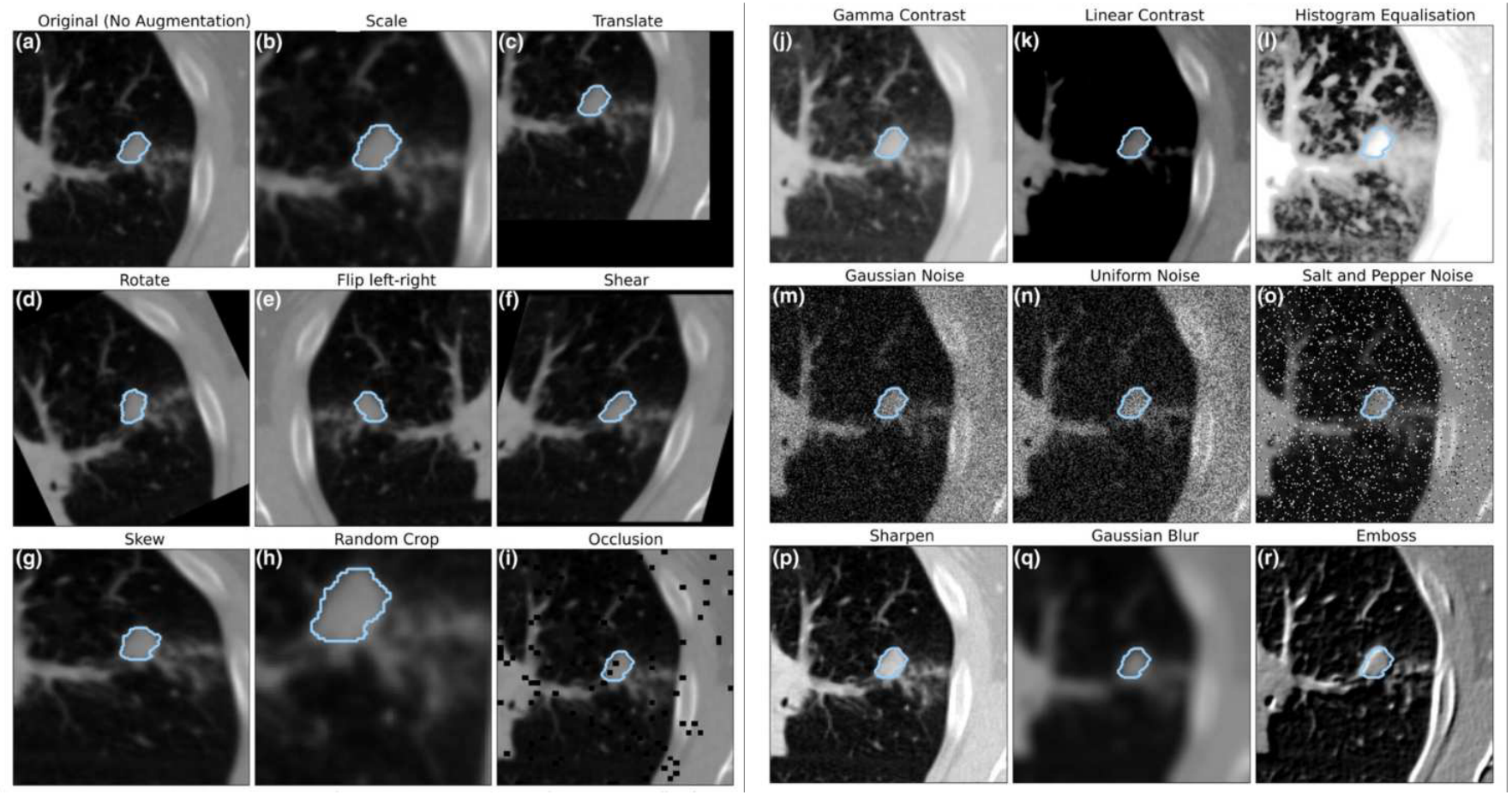
Data Augmentation



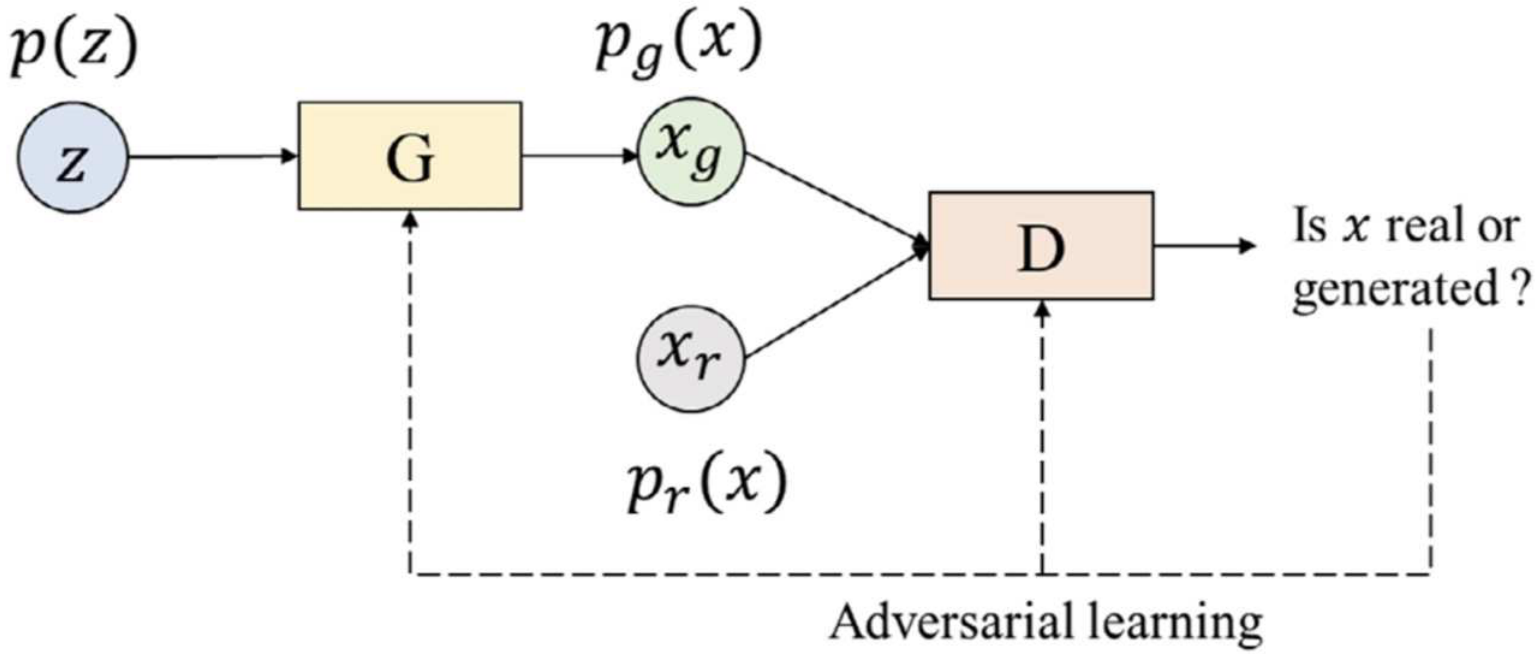
Data Augmentation



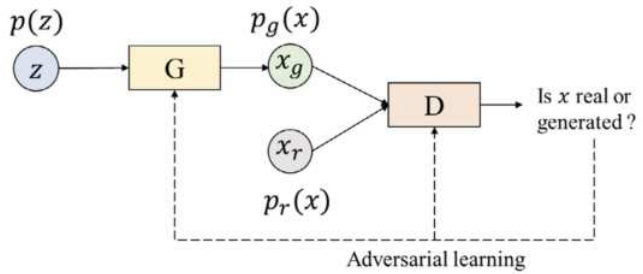
Chlap et al 2021 – Review article

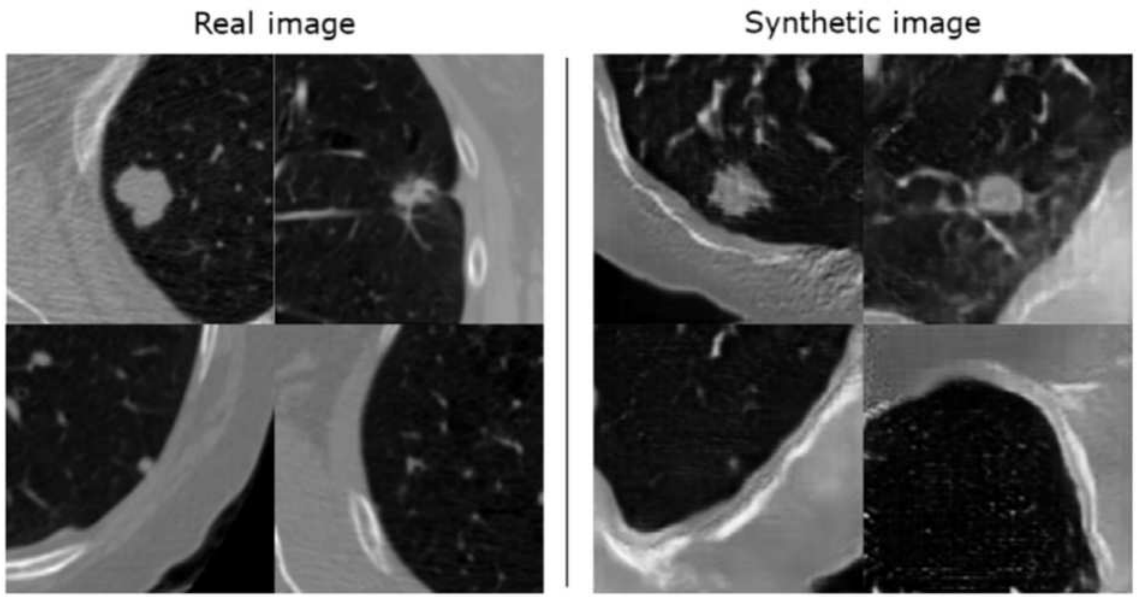


Augmentation via Generative Models (*)



Augmentation via Generative Models (lecture 8)





Data Augmentation - MixUp



60% cat

40% dog

https://arxiv.org/pdf/1710.09412.pdf

Data Augmentation - MixUp



Sample illustration of a Mixup virtual instance comprised of two different classes. In this example, and fMRI scan of non-improver (left) and a fMRI scan of improver (middle) were linearly interpolated to form a third synthetic scan (right) with a soft label of 50% Improver and 50% Non-Improver. (improver –> improve after 1 year treatment). Smuchny et al NeuroImage 2022

🚀 **Key Insight:** MixUp **should NOT be used** when fine-grained texture information is critical (e.g., histopathology images).

🚀 **Pro Tip:** Always test augmented models on **real, unmodified test sets** to avoid learning artifacts.



PART F3 · 2:40–3:00

Pipelines in code: four ways to say the same thing

torchvision v2, Albumentations, Keras preprocessing layers, MONAI. Same concepts, different ergonomics: pick by ecosystem and by how much your task needs masks, boxes, or 3D.

AUTHORED

Written for this course from the libraries' public documentation.

docs.pytorch.org/vision · albumentations.ai/docs · keras.io · docs.monai.io (Project MONAI tutorials)

Two pipelines, one contract

```python
from torchvision.transforms import v2

train_tfm = v2.Compose([
    v2.RandomResizedCrop(224, antialias=True),
    v2.RandomHorizontalFlip(p=0.5),
    v2.ColorJitter(brightness=0.3, contrast=0.3),
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize(mean, std),
])

eval_tfm = v2.Compose([
    v2.Resize(256, antialias=True),
    v2.CenterCrop(224),          # deterministic
    v2.ToDtype(torch.float32, scale=True),
    v2.Normalize(mean, std),      # same numbers
])
```

- **Random at train, deterministic at eval**: the EECS 498 pattern, as code. Validation accuracy measured through random transforms is noise.
- **Normalize appears in both** with identical constants; augmentation appears only on the left.
- Flips are not free everywhere: horizontal flip is wrong for chest X-rays with situs markers, chirality tasks, and text.
- Visual sanity check before training: **plot 8 augmented samples**; if a radiologist would object, so should you.

When labels are pixels too

```python
import albumentations as A

aug = A.Compose([
    A.RandomResizedCrop(size=(512, 512)),
    A.HorizontalFlip(p=0.5),
    A.CLAHE(p=0.3),
    A.RandomBrightnessContrast(p=0.3),
    A.Normalize(mean, std),
])

out = aug(image=img, mask=mask)
# same geometric transform hits both;
# masks are warped with nearest neighbor
```

- **Masks, bounding boxes, and keypoints move with the image**: the reason segmentation and detection work defaults to Albumentations.
- Rich photometric menu (CLAHE, gamma, blur, compression artifacts) that torchvision keeps minimal.
- **Licensing note for the startup side**: classic Albumentations is MIT; the successor AlbumentationsX is AGPL-3.0 / commercial dual-licensed. Check before shipping a product.

Preprocessing that ships inside the model

```
from keras import layers, Sequential

model = Sequential([
    layers.Rescaling(1.0/255),
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.05),
    layers.RandomZoom(0.1),
    # ... the actual network ...
])
# augmentation runs only when training=True;
# at inference the layers pass images through
```

- The train/eval switch is **automatic**: the same exported model is honest at inference time by construction.
- Preprocessing is **saved with the model**, so deployment cannot drift from training: one fewer reproducibility gap.
- Runs on GPU inside the graph; heavy random pipelines stop stealing CPU from the data loader.

Medical pipelines speak DICOM, spacing, and windows

```
from monai.transforms import *

pre = Compose([
    LoadImaged(keys=["image", "label"]),
    EnsureChannelFirstd(keys=["image", "label"]),
    Orientationd(keys=..., axcodes="RAS"),
    Spacingd(keys=..., pixdim=(1.5, 1.5, 2.0),
             mode=("bilinear", "nearest")),
    ScaleIntensityRanged(keys="image",
        a_min=-57, a_max=164,      # CT window (HU)
        b_min=0.0, b_max=1.0, clip=True),
])
```

Pattern from the MONAI spleen segmentation tutorial (github.com/Project-MONAI/tutorials).

- **Dictionary transforms** keep image and label synchronized through every step, in 3D.
- Class 1's threads return as parameters: **voxel spacing** (`Spacingd`) and the **CT window** (`ScaleIntensityRanged` in Hounsfield units).
- Image resampled bilinearly, label with **nearest neighbor**: the segmentation-mask rule from class 1, enforced per key.

Augmenting in 3D, scanner artifacts included

```
aug = Compose([
    RandCropByPosNegLabeld(keys=...,
        spatial_size=(96, 96, 96),
        pos=1, neg=1, num_samples=4),
    RandAffined(keys=..., prob=0.5,
        rotate_range=(0.1,)*3, scale_range=(0.1,)*3),
    RandGaussianNoised(keys="image", prob=0.2),
    RandBiasFieldd(keys="image", prob=0.2),
])

# TorchIO: the scanner-physics menu
import torchio as tio
physics = tio.Compose([tio.RandomMotion(),
    tio.RandomGhosting(), tio.RandomBiasField(),
    tio.RandomElasticDeformation()])
```

- Bias field, ghosting, motion: the artifacts Bagci's slides showed as **problems** become **training signal** when simulated on purpose.
- Patch-based cropping (`RandCropByPosNegLabeld`) balances foreground and background patches: 3D volumes rarely fit in memory whole.
- Keep augmentation **anatomically plausible**: a flipped heart is a diagnosis, not an augmentation.

MixUp and CutMix live after the DataLoader

```
from torchvision.transforms import v2

mixer = v2.RandomChoice([
    v2.MixUp(num_classes=NUM_CLASSES),
    v2.CutMix(num_classes=NUM_CLASSES),
])

for images, targets in loader:
    images, targets = mixer(images, targets)
    # targets are now soft: cat 0.4 / dog 0.6
    loss = criterion(model(images), targets)
```

- These transforms need **pairs of samples**, so they run on the assembled batch, not per image: a different place in the pipeline than everything else today.
- Labels become **soft**; your loss must accept probability targets (cross entropy with label smoothing semantics).
- Johnson's tip: try Cutout and MixUp on small classification sets; Bagci's caution: skip MixUp where fine texture is the diagnosis, and **always evaluate on unmodified images**.

Where each operation lives



offline option: run the blue box once, cache to disk (fast, reproducible, but frozen); keep the amber box on-the-fly so variety stays infinite

Reproducibility checklist:

seed the augmentation RNG (and DataLoader workers) · log every parameter · version the preprocessing code with the model and store the frozen normalization constants next to the checkpoint, never in a notebook cell

If you remember nothing else

- **1.** Classical filters are not obsolete: they are the vocabulary of your first CNN layer and of every artifact you will ever debug.
- **2.** Every preprocessing statistic is fitted on the training split and frozen; medical data adds "and never across a patient boundary."
- **3.** A pretrained backbone is a contract: its resize, its channel order, its normalization constants, or its benefit evaporates.
- **4.** Augmentation is train-time randomness that must be averaged away, or switched off, at test time; evaluation always sees unmodified images.
- **5.** A pipeline nobody can re-run is a result nobody can trust: seeds, parameters, and constants live in versioned code, not in memory.

Your pipeline, end to end

- **Task.** Build one training pipeline for your project data in torchvision v2, Albumentations, or MONAI: deterministic preprocessing, at least three augmentations, honest eval transforms. Plot 8 augmented samples and write the one-paragraph methods description that would let a colleague reproduce it.
- **Leakage audit.** Point at the exact line where your normalization statistics are computed. Which split do they see? Patient-level safe?
- **Reading.** Shorten and Khoshgoftaar, augmentation survey, J. Big Data 2019 · Chlap et al., medical augmentation review, 2021 · MONAI tutorials repo · albumentations.ai/docs · fast.ai Practical Deep Learning (course.fast.ai).
- **Clinic connection.** Session 4's paper clinic question, "could you reproduce the preprocessing from the text alone?", now has teeth: apply it to your own draft.

Source decks used in class 2

1 **EECS 498-007 / 598-005 Deep Learning for Computer Vision, Fall 2020, University of Michigan**, Justin Johnson. Lecture 10, Training Neural Networks, Part 1 (data preprocessing, data augmentation, regularization pattern).

`https://web.eecs.umich.edu/~justincj/teaching/eecs498/FA2020/`

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2 **CS231n Convolutional Neural Networks for Visual Recognition, Spring 2017, Stanford University**, Fei-Fei Li, Justin Johnson, Serena Yeung. Lecture 7, Training Neural Networks, Part 2 (batch normalization recap, transfer learning).

`https://cs231n.stanford.edu/ · slides: cs231n.stanford.edu/slides/2017/`

Slide PDFs © the authors (no explicit license); the companion course notes at cs231n.github.io are MIT-licensed and quoted on the rule slide.

3 **16-385 Computer Vision, Fall 2023, Carnegie Mellon University**, Ioannis Gkioulekas. Lecture 2, Image filtering.

`http://16385.courses.cs.cmu.edu/fall2023/lectures`

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Sources, continued

4 **BME 495 Deep Learning for Biomedical Imaging, Winter 2025, Northwestern University**, Ulas Bagci. Lectures 3–4 (image preprocessing, MRI intensity standardization after Nyul and Udupa) and Lecture 9 (data augmentation; review cited there: Chlap et al. 2021).
`https://bagcilab.com/courses/` · publicly shared by the author; internal reference slides retained at each section's end.

5 **Library documentation** behind the authored code slides: torchvision transforms v2 (docs.pytorch.org/vision), Albumentations (albumentations.ai/docs; AlbumentationsX is AGPL-3.0/commercial), Keras preprocessing layers (keras.io), MONAI and its tutorials (docs.monai.io, github.com/Project-MONAI/tutorials), TorchIO (torchio.readthedocs.io), scikit-image, OpenCV.

6 **Course cross-references**: Kapoor and Narayanan, Leakage and the reproducibility crisis in ML-based science, Patterns 2023 (section 1.11) · nnU-Net, Isensee et al., Nature Methods 2021 (section 3.13) · stain normalization: Reinhard 2001, Macenko 2009, Vahadane 2016 (section 3.13) · OPG-SHAP cropping example (section 3.2).

7 **Compilation note.** Curated by T. Intharah for the Research Master's module Data Preparation for ML and AI, Khon Kaen University, August 2026. For classroom use; each source retains its own license.